

ChE 433 – Process Simulation with ASPEN Plus –Fall 2025

Student Learning Outcome 2

Rubric for assessment Assignments #1, #2, #3

Assignment #1: Fundamentals of Process Simulation and Cost Estimation.

Assignment #2: Design and Optimization of a Multistage Refrigeration Loop

Assignment #3: Simulation of a Biomass Pyrolysis Unit

SLO 2 Performance Indicator (PI)	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1. Developing a design strategy Assignment #2 100 pts	No design strategy; haphazard approach (<70 pts)	Uses a design strategy with guidance (70-90 pts)	Develops a design strategy, decomposition of work into subtasks, development of a timetable (>90 pts)
2. Applying engineering and/or scientific principles correctly to design practical processes Assignment #3 100 pts	No application of engineering and/or scientific principles (<70 pts)	Includes only minor or cursory consideration of economic, safety, and environmental constraints (70-90 pts)	Applies engineering and/or scientific principles correctly to design practical processes (>90 pts)
5. Supporting design procedure with documentation and references Assignment #1 100 pts	Design is done incompletely without the proper equations and without references (<70 pts)	Minimal or incorrect use of computer tools and engineering resources (70-90 pts)	Understands how areas interrelate and demonstrates ability to integrate prior knowledge into a new problem (>90 pts)

Rubric					
PI	Problem	Total pts	Unsatisfactory	Developing	Satisfactory
1	1[Assign #2]	100	<70	70-90	>90
2	2[Assign #3]	100	<70	70-90	>90
5	5[Assign #1]	100	<70	70-90	>90

RESULTS BY PERFORMANCE INDICATOR			
Rubric #	1	2	5
Mapped SLO	2	2	2
Average	2.63	2.26	2.58
Standard Deviation	0.68	0.65	0.61
S	14	7	12
D	3	10	6
U	2	2	1
Checksum	19	19	19

RESULTS BY STUDENT LEARNING OUTCOME			
SLO	2		
Average Score	2.49		
Standard Deviation	0.16		

Analysis of Results

Average scores were fairly uniform although there was some variation among students. More students were strong in applying engineering and/or scientific principles correctly to design practical processes. Overall performance was satisfactory for all performance indicators despite the very stringent criteria used for assessment. Assignment #3 for Student Learning Outcome 5, was a bit challenging for a large number of students due to its complexity both in the reactor modeling and the downstream processing of the pyrolysis reactor effluent. The simulation also involved solids handling which was a new unit introduced to the students.

Assignment #1: Design and Cost Analysis of a Hydrogenation Unit

Your first assignment after six months of training in the **UIC Specialty Chemicals Corporation** is to provide a material balance for a cyclohexane production unit and preliminary cost estimates of the utilities. A simple process flow diagram (shown in Figure A below) has been given to you by a mid-level engineer who has decided to leave the company. The data for this process are given below:

Benzene (BZ) Feed

Flow: 92.14 lbmole/hr

Temperature: 100 °F

Pressure: 15 psia

Hydrogen (H₂) Feed

Flow: 283.8 lbmole/hr

Temperature: 120 °F

Pressure: 335 psia

H₂ mole fraction : 0.975

N₂ mole fraction : 0.005

CH₄ mole fraction : 0.020

Charge Heater

Temperature: 300 °F

Pressure: 330 psia

Reactor

Temperature: 400 °F

Pressure: 315 psia

Benzene Conversion: 99.86%

H₂ to BZ molar ratio: 3.3

High Pressure (HP) Product Separator (Flash Drum)

Temperature: 120 °F

Pressure : 300 psia

Low Pressure (LP) Product Separator (Flash Drum):

Pressure : 15 psia

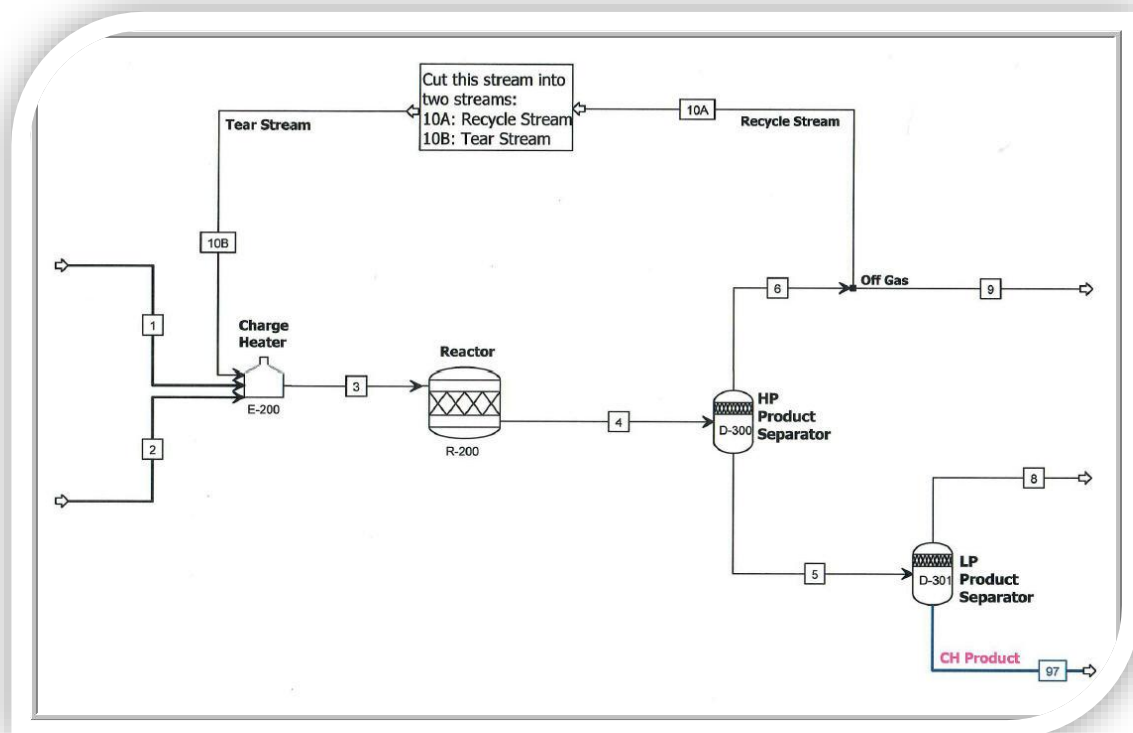
Your tasks

1. Complete the process flowsheet by adding the missing unit operations and provide a brief explanation for every unit you add.
2. Prepare the material balance in the format of the **ENG Calculations** of the **CHE Calculator**. Provide a brief explanation on how you calculate the effluent streams from each unit operation. Additionally provide a brief description of the numerical method(s) you have used to derive the material balance
3. Provide a preliminary cost of all utilities
4. Propose a method to reduce the utilities cost and do a simulation to calculate the cost reduction of your proposal

Data and Assumptions:

1. You may assume that the hydrocarbon mixtures with hydrogen presence in the process follow the Peng-Robinson equation of state
2. You may also assume that the K-values for the flash calculations can be calculated from Raoult's law or the Wilson equation
3. The fuel in the charge heater is natural gas
4. Utility cost data can be obtained from a variety of sources

Figure A: Hydrogenation of Benzene to Cyclohexane



Assignment #2: Design and Optimization of a refrigeration loop

Methane at temperature 25 °C and pressure 30 bar is to be precooled down to -38 °C prior to liquefaction. The precooling will be done using propane as the refrigerant. Design a multi-stage refrigeration loop and calculate the annual operating cost and the capital cost. You may attempt to minimize the total annualized cost for your design. The flow rate of methane is 250 MMSCFD. The isentropic efficiency of the compressors can be taken as 80% and the mechanical efficiency equal to 95%.

Notes:

For both problems assume that the physical properties of methane and propane are best described by the Peng-Robinson Equation of State

Minimum temperature approach for condenser may be taken as $\Delta T_{\min} = 10\text{ }^{\circ}\text{C}$

Minimum temperature approach for evaporators:

Stage	$\Delta T_{\min}$
1	8
2	6
3	4
4	2

Gavin Towler and Ray Sinnott, **Chemical Engineering Design**. Principles Practice and Economics of Plant and Process Design, Second Edition © Elsevier Ltd. 2013.

Data:

Heat Exchangers: Shell and Tube Single Pass

Purchase cost of the Heat Exchanger (\$ January 2010): $C_{\text{PURCHASE}} = 28,000 + 54 A^{1.2}$ $A[=]m^2$

Heat Transfer Area: A

Lang Factor: $LF = 3.5$

Compressors: Centrifugal

Purchase cost of the Compressor (\$ January 2010): $C_{\text{PURCHASE}} = 580,000 + 20,000 P^{0.6}$ $P[=]kW$

Power Requirement: P

Lang Factor: $LF = 2.5$

Valves: Expansion Valves

Purchase cost of the Valve (\$ January 2010): $C_{\text{PURCHASE}} = 8,142 m^{0.7}$ $m[=]kg / s$

Mass Flow Rate: m

Lang Factor: $LF = 1.3$

Drums: Flash Drum with Hemispherical Head and Bottom

Purchase cost of the Drum (\$ January 2010): $C_{\text{PURCHASE}} = 11,600 + 34 W^{0.85}$ $W[=]kg$

Shell Mass: W

Lang Factor: $LF = 4.0$

Data (Continued):

Annualized Capital Charge Ratio: $ACCR = 0.2$

Annualized Capital Cost (\$/year): $ACC = C_{PURCHASE} * LF * ACCR$

Cost of Cooling Water: $C_{CW} = \$0.354 / GJ$

Cost of Electricity: $C_{EL} = \$0.06 / kWh$

Overall Heat Transfer Coefficient Condenser: $U = 425.6 \text{ W} / (m^2 \cdot K)$

Overall Heat Transfer Coefficient Evaporators: $U = 425.6 \text{ W} / (m^2 \cdot K)$

Heat Capacity of Water: $C_p = 4.184 \text{ kJ} / (kg \cdot K)$

Cooling Water Temperature: $T_{CWin} = 25^\circ C$

Cooling Water Pressure: $P_{CWin} = 20 \text{ psia}$

Cooling Water Temperature Rise: $10^\circ C$

Operating Year: 8000 hrs

Cost Escalation Factor: 1.50

Additional Instructions for your Report:

Your Custom table should contain three tabs

Tab 1 (Name KPIs): Display COP, Total Power(kW), Total Annualized Capital Cost (MM\$/oper_year), Total Utility Cost (MM\$/oper_year) and the sum of the two costs

Tab2 (Name: Detailed Cost): The installed cost (\$) of each equipment and cost of each utility in \$/oper_year

Tab3 (Name: Optimal Pressure): The final discharge pressure of each stage in bar

Assignment #3: Biomass Pyrolysis Simulation

Biomass conversion processes that are based on pyrolysis offer high yields of liquid transportation fuels such as gasoline and diesel blend stocks, or refinery intermediates. Fast pyrolysis proceeds by heating the biomass to an intermediate temperature where volatilization begins but the gasification of vapor-phase products does not occur. Fast pyrolysis generates a liquid bio-oil that can be transported efficiently and upgraded to fuels.

In a conceptual process design (see Process Flowsheet Diagram Figure 1. below), the fast pyrolysis reactor system is usually represented as a dual circulating fluidized bed. Biomass and hot sand are introduced into the bottom of a vertical riser reactor. As the solids are carried up the riser by a fluidizing gas, the rapid heating of biomass particles to around 450-500 degrees Celsius, produces non-condensable gases and vapors of oxygenated hydrocarbons. Unreacted biomass flows out as solid char. Solid and vapor phases exit the top of the riser and are separated in a cyclone. Sand and char drop into a char combustor, which is basically a second fluidized bed where the unreacted char is combusted in air to reheat the sand. The energy to drive the endothermic pyrolysis reactions is thus provided via the hot sand, by partial combustion of the biomass itself. The vapor phase is condensed downstream to form the bio-oil. In some cases, the bio-oil separates into organic and aqueous phases. Fast pyrolysis bio-oils can be as high as 50% oxygen by mass and are generally assumed to be hydro-treated downstream, to reduce oxygen.

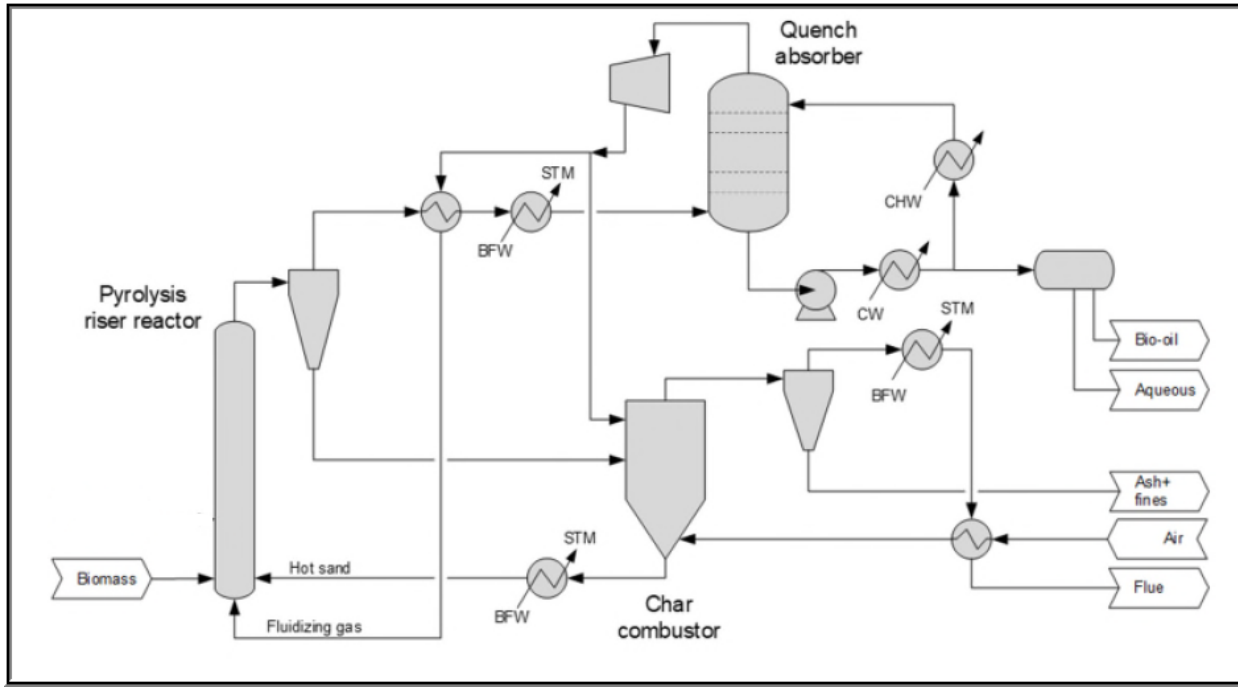


Figure 1: Process Flow Diagram(PFD) of Biomass Fast Pyrolysis

The objective of the assignment is to prepare an ASPEN Plus simulation and calculate:

1. Total Power(KW) consumed
2. Total High(250 °C) and Low(125 °C) pressure steam generated (MMSCFD)
3. Mass yield (%) of incoming carbon to **Bio-oil**
4. Mass yield (%) of incoming carbon to **Flue Gas**
5. Mass yield (%) of incoming carbon to **Waste Water**

Your final report should also include the following

1. Material and Energy Balance with the UIC template
2. A brief outline of your methodology for modeling the pyrolysis reactor
3. A brief outline of your methodology for modeling the char combustor

The Quench Absorber can be simulated using a SEP2 Aspen Plus Unit Operation block

Data for the simulation

1. Bone dry biomass flow rate: **50,000 kg/hr**
2. Biomass feed temperature: **25 deg C**
3. Biomass feed pressure: **2.3 atm**
4. Moisture weight fraction: **0.1**
5. Fluidization gas to Sand mass ratio: **0.1**
6. Bleed Gas to Combustor: **13,000 kg/hr**
7. Bleed Gas temperature: **80 deg C**
8. Bleed Gas pressure: **2.31 atm**
9. Sand to dry biomass mass ratio: **7.0**
10. Biomass **PROXANAL**
 - a. Moisture:
 - b. Fixed Carbon (FC): $0.85 \cdot (100 - \text{ASH})$
 - c. Volatile Matter (VM): $0.15 \cdot (100 - \text{ASH})$
 - d. Ash (ASH): 0.3000
11. Biomass **ULTANAL**
 - e. ASH: 0.30
 - f. CARBON: 49.66
 - g. HYDROGEN: 6.31
 - h. NITROGEN: 0.10
 - i. CHLORINE: 0.00
 - j. SULFUR: 0.08
 - k. OXYGEN: 43.55
12. Biomass **SULFANAL**
 - l. PYRITIC: 0.00
 - m. SULFATE: 0.04
 - n. ORGANIC: 0.04

13. Pyrolysis Reactor Pressure Drop: 0.3 atm
14. Pyrolysis Reactor Temperature: 480 deg C
15. Pyrolysis Reactor Yield (from Experiments)

o. Gas:	12.9%
i. H2:	0.10%
ii. CO:	5.82%
iii. CO2:	6.03%
iv. CH4:	0.53%
v. C2H4:	0.42%
p. Water:	14.8%
q. Char:	9.7%
r. Organics:	62.6%
i. Acids[Acetic Acid]:	11.1%
ii. Carbonyls[Acetone]:	12.7%
iii. Phenols[m-Cresol]:	4.0%
iv. Methoxyphenols[Guaiacol]:	27.5%
v. Levoglucosan:	4.4%
vi. Furfural:	2.9%

16. Combustor Pressure: **1.12 atm**
17. Combustion products: **NO2, SO2, CO2, H2O**
18. Oxygen in the combustor effluent: **2% mol**
19. Fluidization Gas temperature: **375 deg C**
20. Fluidization Gas pressure: **2.31 atm**
21. Fluidization Gas and Bleed Gas composition, **mol%**
 s. CO: 31.19
 t. CO2: 20.11
 u. Water: 15.09
 v. NO2: 0.30
 w. SO2: 0.91
 x. Methane: 2.62
 y. Ethylene: 1.51
 z. Acetone: 28.27
22. Sand component: SiO2
23. Temperature of the Quench Absorber (SEP2): 45 deg C
24. Pressure of the Quench Absorber: 1.7 atm

Problem A: Review of Chemical Engineering Fundamentals

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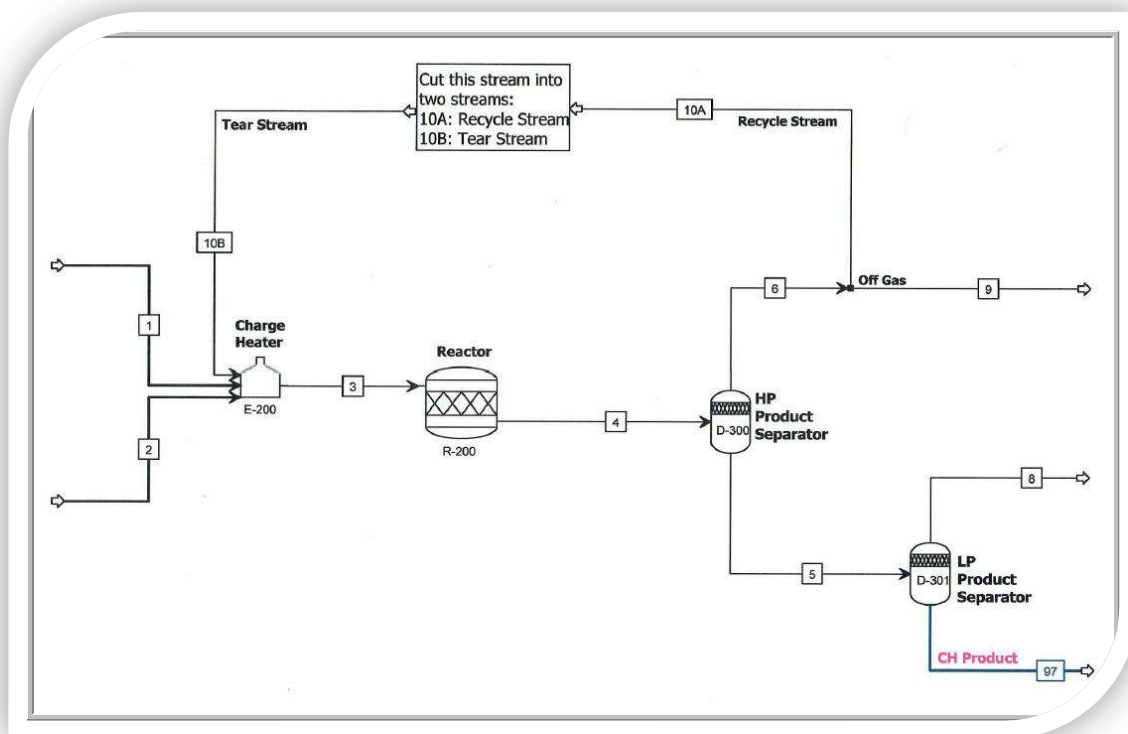
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Data and Assumptions:

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2. You may also assume that the K-values for the flash calculations can be calculated from Raoult's law or the Wilson equation
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Figure A: Hydrogenation of Benzene to Cyclohexane



Introduction

Cyclohexane is a key intermediate for nylon and solvent manufacture and is commonly produced by the catalytic hydrogenation of benzene. In this assignment, we evaluate a streamlined process in which benzene and hydrogen are brought to reaction conditions and converted as follows:



The reaction mixture is then cooled and separated to recover the liquid product while recycling the hydrogen-rich gas. Our objectives are to complete the process flowsheet, perform material balances, estimate utility costs, and simulate a practical way to lower those costs. Most calculations will be performed using the **CHE Calculator** provided by Dr. Michael Caracotsios.

1. Process Flowsheet

The base PFD is mostly complete, but a few additions are required to ensure stable operation and clean separations. These edits focus on conditioning the combined feed, establishing clear pressure/level control points, and cooling the reactor effluent so the flash drums perform as specified. In summary, a mixer, three throttling valves, and a condenser are added. The annotated flowsheet image and the table below summarize the changes and their purpose.

Figure 1: Process flowsheet

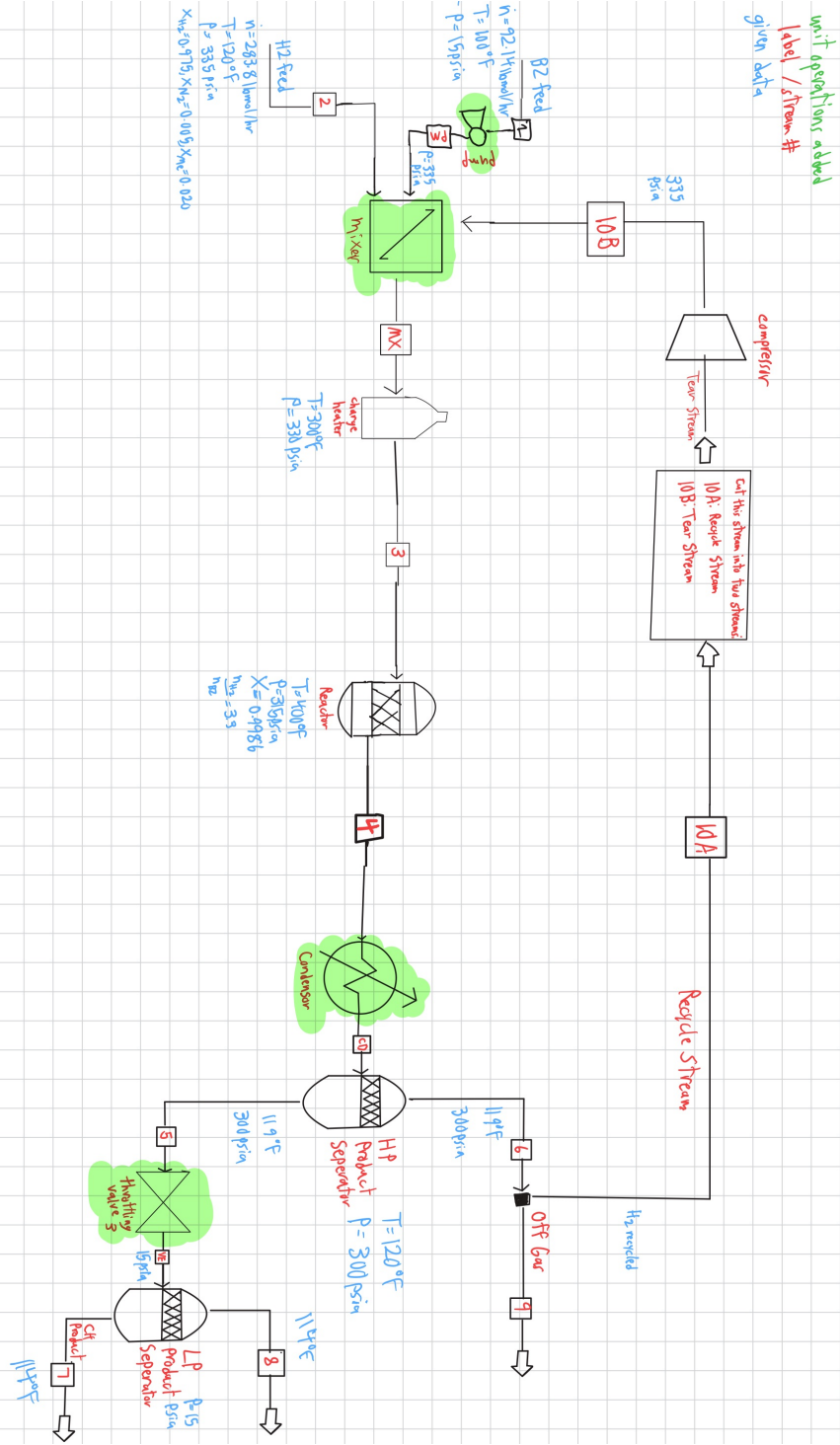


Table 1: Unit operations added

Unit operation	Location	Label	Explanation
Mixer	Beginning of flowsheet	MX	Combines benzene, hydrogen, and recycle into one uniform feed so the H_2/BZ ratio is consistent and the heater/reactor see steady conditions.
Pump	After BZ feed	PM	to raise the BZ stream pressure so it matches the recycle and H_2 feeds at the mixer; this ensures positive, stable flow into the process and prevents backflow or flashing in upstream lines.
Condenser	After Valve 2	CD	Cools the hot reactor effluent to near the HP separator temperature so a liquid phase forms and the flash separation is effective.
Compressor	After Tear Stream	10B	on the recycle line before the mixer to lift the recycle vapor back to the mixer setpoint; this guarantees a reliable recycle, maintains the H_2/BZ ratio with less fresh H_2
Valve	HP separator bottoms to LP separator	VE	Level-control/let-down valve that maintains HP drum level and drops pressure to the LP flash for additional separation.

What if these changes were omitted? Without the mixer, the H_2 /benzene ratio would wander as streams meet, giving unstable feed conditions and variable conversion. Without the condenser, the HP separator would receive mostly hot vapor, produce little liquid product, and the recycle composition would drift. Without the HP bottoms let-down/level valve, The HP→LP pressure drop would occur in the line, causing premature flashing. The LP drum could not be held at 15 psia, and separation would degrade. Without the benzene feed pump, the BZ stream would be under-pressured relative to recycle and H_2 , causing poor mixing, flow reversal risks, and feed starvation. Without the recycle compressor, the recycle flow would collapse or become erratic, forcing higher fresh H_2 usage and making the mixer (and reactor) pressure harder to hold. Overall, product recovery and controllability would degrade, raising utility use and off-spec risk.

2. Material Balance

We received a flow diagram detailing the inputs to the cyclohexane production system, specifically for Stream 1 and Stream 2, each with distinct flow rates and compositions. All units obey an overall mole balance (*moles in* = *moles out* $\pm$ reaction), and component molar flows were computed as $\dot{n}_i = z_i \dot{n}_{\text{tot}}$, where z_i is the mole fraction and $\dot{n}_{\text{tot}}$ the total molar flow. The core challenge was coordinating the recycle stream to determine how much of each component is returned to the reactor while satisfying the design constraint of a hydrogen-to-benzene (H_2/BZ) molar ratio of 3.3.

Our approach began by temporarily ignoring the recycle stream. This allowed us to assign placeholder values to unknown flow rates and carry out forward calculations. Next, we estimated the *reactor outlet* using the benzene hydrogenation stoichiometry $\text{BZ} + 3\text{H}_2 \rightarrow \text{CH}$. Let $\xi = X_{\text{BZ}} \dot{n}_{\text{BZ},\text{in}}$ denote the extent (moles of benzene reacted). Then $\dot{n}_{\text{BZ},\text{out}} = \dot{n}_{\text{BZ},\text{in}} - \xi$, $\dot{n}_{\text{H}_2,\text{out}} = \dot{n}_{\text{H}_2,\text{in}} - 3\xi$, and $\dot{n}_{\text{CH},\text{out}} = \dot{n}_{\text{CH},\text{in}} + \xi$, while inerts (CH_4 , N_2) were unchanged across the reactor. Once we reached the high-pressure (HP) product separator, we performed vapor–liquid equilibrium (VLE) calculations to account for phase behavior (see table “HP drum”). The VLE vapor fraction α from the Rachford–Rice calculation set the phase split: the vapor effluent molar flow is $\dot{n}_v = \alpha \dot{n}_{\text{feed}}$, and the liquid effluent is $\dot{n}_\ell = (1 - \alpha) \dot{n}_{\text{feed}}$.

To initialize the system, we made an educated guess for the purge fraction of the recycle, which provided a placeholder value for Stream 10A. With all streams now defined by placeholders, we copied the values for Stream 10A into Stream 10B, solved Rachford–Rice once for an initial α , and then applied Newton’s method to iteratively converge the correct flow rates for all streams; compositions followed from these flows.

In our formulation, the left-hand side (LHS) of the convergence equations included the recycle flow rates, the Rachford–Rice (RR) equation, and the H_2/BZ ratio; the right-hand side (RHS) consisted of the tear-stream flow rates, the target RR, and the target H_2/BZ ratio. By adjusting the vapor fraction, purge fraction, and the tear stream, we obtained values that satisfy the required H_2/BZ ratio of 3.3 while meeting the RR relation and the overall mass balance. All Material Balance results appear in the first table of Figure 2.

Similarly, once the recycle flow was established, we computed the correct vapor fraction in the LP product separator and determined the compositions and flow rates in the downstream streams. All calculations are shown in Figure 2, and all pressures, temperatures, compositions, and enthalpies are summarized in Figure 3.

ENG Calculations														
1	2	PM	108	MX	3	4	CD	5	VE	7(top)	8(top)	9	10A	5
42	Molar flow for comp (lbmole/hr)													
43	CH6	92.1400	0.0000	92.1400	0.0051	92.1451	92.1451	0.1289	0.1238	0.1289	0.1212	0.0026		purge fraction
44	H2	0.0000	276.7050	0.0000	27.3739	304.0789	28.0305	27.3081	0.1244	0.1246	0.0008	0.1217		0.0191
45	CH4	0.0000	5.6760	0.0000	107.9656	113.6416	113.6416	110.0726	3.5689	0.5520	0.0633	3.5056		H2/R2 ratio
46	N2	0.0000	1.4190	0.0000	44.3019	45.7209	45.7209	45.1665	0.5544	3.5596	0.0262	0.5282		3.3000
47	CH2	0.0000	0.0000	0.0000	3.6037	3.6037	95.6198	3.6740	0.0793	3.6037	91.9458	90.1614	1.7844	desired
48														
49	Total	92.1400	283.8000	92.1400	183.2502	559.1902	283.1419	186.8285	3.5763	183.2502	96.3154	90.3729	5.9425	
50														
51	Mixer Inlet flow													
52	CH6	92.1400	0.0000	0.0051	92.1451	0.1648								
53	H2	0.0000	276.7050	27.3739	304.0789	0.5438								
54	CH4	0.0000	5.6760	107.9656	113.6416	0.2032								
55	N2	0.0000	1.4190	44.3019	45.7209	0.0818								
56	CH2	0.0000	0.0000	3.6037	3.6037	0.0064								
57					559.1902	1.0000								
58														
59	Conversion	0.9985												
60	extent	92.0161												
61														
62	HP drum													
63	CH6	0.0005	0.001285	0.0000	0.1494	0.0217	RR							
64	H2	0.0990	0.001271	0.1494	117.5000	-0.1481	T	48.8889						
65	CH4	0.4014	0.0371	0.5882	15.9000	-0.5521	P	20.6842						
66	N2	0.1615	0.0058	0.2418	42.0000	-0.2360	F	288.1419						
67	CH2	0.3377	0.0546	0.0197	0.0206	0.9350	L	96.3154						
68		1.0000	1.0000	1.0000			V	186.8285						
69														
70														
71	VLE for HP separator													
72	RR	0.000000	0											
73	Alpha	0.6588												
74														
75														
76	LP drum													
77	CH6	0.0013	0.0013	0.0004	0.3214	0.0009	RR							
78	H2	0.0013	0.0000	0.0205	2461.0000	-0.0205	T	32.3379						
79	N2	0.0371	0.0007	0.5889	842.0000	-0.5892	P	1.0342						
80	CH4	0.0058	0.0003	0.0889	306.0000	-0.0886	F	96.3154						
81	CH2	0.5546	0.3977	0.3003	0.3010	0.6974	L	90.373115						
82							V	5.94245						
83	VLE for LP separator													
84	RR	0.0000												
85	Alpha	0.6617												
86														
87														
88														

Figure 2: Additional Calculations

3. Utilities Cost Analysis

We estimated utilities from stream duties and powers computed in the CHE Calculator. For each equipment item, the duty Q [kJ/s] is the net *enthalpy flow* difference between outlet and inlet streams using the Calculator's `Enthalpy(...)` function (inputs: composition, T , P , and Z). Entropies were obtained with the matching `Entropy(...)` function, and the compressibility factor Z was also generated by the Calculator. The compressor outlet temperature was found by an *isentropic* balance: we goal-seeked $\Delta s = 0$ by adjusting the compressor effluent T . *For costing, we used efficiencies provided by Dr. Caracotsios: $\eta_{\text{pump}} = 0.70$ and $\eta_{\text{comp}} = 0.80$. using enthalpies from the CHE Calculator; the electrical duty used for costing was $W_{\text{real}} = W_s/\eta_c$.*

The valve outlet temperature was found from an *isenthalpic* balance ($\Delta h = 0$). The HP drum effluent temperature was assumed to drop only slightly (from 120°F to 119°F). The mixer outlet temperature was approximated by a mass-weighted average of its inlets (Streams 1, 2, and 10B): $T_{\text{mix}} = \sum_i w_i T_i$, with $w_i = \dot{m}_i / \sum \dot{m}_i$. All other temperatures and pressures were taken from the problem statement, assumed unchanged from the previous stream, or set to the values provided by Dr. Caracotsios (e.g., condenser and LP drum effluents).

Only *utility costs* to operate the equipment for one year were included; capital and raw materials were excluded. Electricity for the compressor and pump was priced at 0.06 \$/kWh: since $1 \text{ kJ/s} = 1 \text{ kW}$,

$$C_{\text{elec}} = \left(\sum W_i [\text{kW}] \right) \times 8000 [\text{h/yr}] \times 0.06 [\$/\text{kWh}].$$

Thermal utilities were priced from duties in kJ/s converted to GJ/yr using 28,800,000 s/yr and 10^6 kJ/GJ :

$$C_{\text{CW}} = |Q_{\text{cool}}| \times \frac{28,800,000}{10^6} \times 0.354, \quad C_{\text{NG}} = Q_{\text{heat}} \times \frac{28,800,000}{10^6} \times 6.0.$$

for cooling water and natural gas respectively. The total annual utility cost is the sum over the four contributors (heater, condenser, compressor, pump):

$$C_{\text{total}} = C_{\text{NG}} + C_{\text{CW}} + C_{\text{elec}}.$$

Using the calculated duties/powers, the annual utility costs are:

$$\text{Pump} = \$2,215.74 \quad \text{Compressor} = \$4,05.49, \quad \text{Charge Heater} = \$57,792.74, \quad \text{Condenser} = \$4,502.89,$$

so the total is:

$$C_{\text{total}} = \$68,561.87.$$

All intermediate values and calculations are shown in Figure 3. s/yr and h/yr values were provided by Dr. Caracotsios.

Eng Calculations		Perry-Robinson													
Molar Composition	1	2	PM	10B	MX	3	4	CD	6	9	10A	5	VE	7	8
CH6	1.0000	0.0000	1.0000	0.0000279	0.1648	0.1648	0.0005	0.0005	0.0000	0.0000	0.0000279	0.0013	0.0013	0.0013	0.0004
H2	0.0000	0.9750	0.0000	0.1493797	0.5438	0.5438	0.10	0.0990	0.1494	0.1494	0.1494	0.0013	0.0013	0.0000	0.0205
CH4	0.0000	0.0200	0.0000	0.1589703	0.2032	0.2032	0.4014	0.4014	0.5892	0.5892	0.0371	0.0057	0.0057	0.0007	0.5899
N2	0.0000	0.0050	0.0000	0.2417566	0.0818	0.0818	0.1615	0.1615	0.2418	0.2418	0.2418	0.0058	0.0370	0.0003	0.0889
CSH12	0.0000	0.0000	0.0000	0.0196654	0.0054	0.0054	0.3377	0.3377	0.0197	0.0197	0.0197	0.9546	0.9547	0.9977	0.3003
Sum	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Stream Properties	1	2	PM	10B	MX	3	4	CD	6	9	10A	5	VE	7(bottom)	8(top)
Molar Flow, kmol/hr	92.1400	283.8000	92.140000	183.2502	559.1902	559.1902	283.1419	283.142	187	3.5763	183.2502	96.315359839	96.3154	90.3731	5.9422
Molar Flow, kmol/s	0.0116	0.0358	0.0116	0.0231	0.0705	0.0705	0.0357	0.0357	0.0235	0.0005	0.0231	0.0121	0.0121	0.0114	0.0007
Volume Flow, m ³ /s															
Temperature, C	37.7778	48.8889	38.9641	56.9965	44.9233	148.8889	204.4444	48.8889	48.3333	48.3333	48.3333	48.3333	32.3379	45.5556	45.5556
Pressure, bar	1.0342	23.0974	23.0974	23.0974	23.0974	22.7526	21.7184	20.6842	20.6842	20.6842	20.6842	20.6842	1.0342	1.0342	1.0342
Vapor Fraction															
Molecular Weight, kg/kmol				compressor out							compressor in				
Molar Enthalpy, kJ/kmol	57682.037	-821.987	52710.378	-45987.535	-1973.156	2768.708	-59811.930	-71682.167	-45982.934	-45982.934	-45987.535	-147619.537	-147619.34	-151875.2	-80031.592
Vapor Phase															
Liquid Phase															
Molar Entropy, kJ/(kmol K)	-244.614	-24.401	-244.347	-72.292	-60.032	-47.034	-190.354	-221.104	-72.292	-72.292	-72.292	-576.530	-581.156	-599.737	-192.114
Vapor Phase															
Liquid Phase															
Molar Density, kmol/m ³															
Enthalpy Flow, kJ/s	601.74	29.39	604.97	-4054.854	-159.37	165.67	-2115.57	-2357.64	-1082.43	-20.72	-1081.7063	-1791.44	-1791.44	-1729.38	-59.92
Entropy Flow, kJ/s K	-2.94	-0.87	-2.94	-1.67	-4.23	-3.37	-6.79	-7.89	-1.70	-0.03	-1.67	-7.00	-7.05	-6.83	-0.14
Availability, kJ/s				Compressor	Heater		Condenser								
Duty (Power), kJ/s [kW]	3.23			6.338	334.45		-44.67								
Efficiency	0.70			0.80											
Real Power	4.6161			8.4385											
Yearly Cost(\$)	39628.9499			67508.2142	963272361.7	9632.124	-12720039065	-17220.039							
	\$2,215.74			\$ 4,050.49	\$57,792.74		\$4,502.89		\$68,561.87						
Compressor				compressor	Charge Heater		Condensor								
Compressibility Factor	0.0035	1.0064	0.0786	0.9782	0.9497	0.9623	0.9498	0.8718	0.9780	0.9780	0.9780	0.0892	0.0042	0.0041	0.9915

Figure 3: Utility duty and cost worksheet

4. Utilities Cost Reduction

We implemented a closed hot-water loop that circulates around the reactor to pick up sensible heat, then passes through a heat exchanger placed after the recycle compressor to warm Stream 10B to about 80°C. This raises the mixer inlet (and thus outlet) temperature; a warmer feed requires a smaller temperature change across the heater, thus decreasing net enthalpy flow.

That drop in duty directly reduces the natural-gas cost for the charge heater and may slightly ease condenser duty as well. This modification does not add meaningful operating cost because it recovers heat already available near the reactor (no extra fuel or purchased cooling), and the only incremental utility is a small circulation-pump load, which we ignore since its cost is negligible relative to the savings.

New annual utility costs: Pump = \$2,215.74; Compressor = \$4,050.49; Charge heater = \$54,207.35; Condenser = \$4,502.89; Total = \$64,976.47; Savings vs. base = \$3,585.40.

Figure 4 shows the revised calculator sheet; the only difference from Figure 3 is the updated mixer outlet temperature reflecting the warmed recycle.

ENG Calculations													Peng-Robinson													
1	Molar Composition				1		2		PM		10B		MX		3		4		CD		6		9		10B	
2	C6H6				1.0000		0.0000		1.0000		0.0000279		0.1648		0.1648		0.0005		0.0005		0.0000		0.0000		0.0000	
3	H2				0.0000		0.9750		0.0000		0.1493797		0.5438		0.5438		0.10		0.0990		0.1494		0.1494		0.0000	
4	CH4				0.0000		0.0200		0.0000		0.5891703		0.2032		0.2032		0.4014		0.4014		0.5892		0.5892		0.0000	
5	N2				0.0000		0.0050		0.0000		0.2417566		0.0818		0.0818		0.1615		0.1615		0.2418		0.2418		0.0000	
6	C6H12				0.0000		0.0000		0.0000		0.0196654		0.0064		0.0064		0.3317		0.3317		0.0197		0.0197		0.0000	
7	Sum				1.0000		1.0000		1.0000		1.0000		1.0000		1.0000		1.0000		1.0000		1.0000		1.0000		1.0000	
8																										
9																										
10																										
11																										
12																										
13	Stream Properties				1		2		PM		10B		MX		3		4		CD		6		9		10B	
14	Molar Flow, lbmol/hr				92.1400		283.8000		92.140000		183.2502		559.1902		559.1902		283.1419		283.142		187		3.5763		183	
15	MolarFlow, kmols				0.0116		0.0358		0.0116		0.0231		0.0705		0.0705		0.0357		0.0357		0.0235		0.0005		0	
16	Volume Flow, m³/s								38.9641		56.9965		51.7618		148.8889		204.4444		48.8889		48.3333		48.3333		48	
17	Temperature, C				37.7778		48.8889		37.7778		56.9965		51.7618		148.8889		204.4444		48.8889		48.3333		48.3333		48	
18	Pressure, bar				1.0342		23.0974		23.0974		23.0974		23.0974		22.7526		21.7184		20.6842		20.6842		20.6842		20	
19	Vapor Fraction																				1					
20	Molecular Weight, kg/kmol																									
21											compressor out														compressor in	
22	Molar Enthalpy, kJ/kmol				51832.037		-821.987		62110.370		-45650.663		-1693.666		2768.708		-59311.930		-71692.157		-45982.934		-45982.934		-45982.934	
23	Vapor Phase																									
24	Liquid Phase																									
25	Molar Entropy, kJ/(kmol K)				-244.614		-24.401		-244.347		-72.292		-59.116		-47.034		-190.354		-221.104		-72.292		-72.292		-7	
26	Vapor Phase																									
27	Liquid Phase																									
28	Molar Density, kmol/m³																									
29																										
30	Enthalpy Flow, kJ/s				6011.74		-29.39		604.97		-1054.9564		-118.63		195.07		-2115.97		-2557.54		-1082.43		-20.72		-1051	
31	Entropy Flow, kJ/s K				-2.84		-0.87		-2.84		-1.67		-4.17		-3.31		-6.79		-7.89		-1.70		-0.03			
32	Availability, kJ/s				Pump						Compressor		Heater				Condensor									
33	Duty [Power], kJ/s [kW]				3.23						6.7306		313.70				-441.67									
34	Efficiency				0.70						0.80															
35	real power				4.6161						8.4385															
36	kWhr				36928.9499						67508.2142		9034557552.8		9034.558		-12720039065		-12720.039		total cost					
37	Yearly Cost(\$)				\$2,216.74						\$ 4,050.49		compressor		Charge Heater		Condensor		\$4,502.89		\$64,976.47		total cost			
38					pump						compressor															
39	Compressibility Factor				0.0035		1.0064		0.0786		0.9782		0.9537		0.9823		0.9488		0.8118		0.9780		0.9780		0	
40																										

Problem A: Optimal Cooling Water Flow Rate in Condenser

Your engineering team has been asked to design a single-pass condenser for the condensation of propane vapor at 20 bar pressure. The vapor is at its dew point, and the flow rate is 2088 kmol/hr. Cooling water at 25 °C is available from a nearby cooling tower. Estimate the optimal water circulation rate in kg/s.

Data:

Heat Exchanger Type: Shell and Tube Single Pass

Purchase cost of the Heat Exchanger (\$ January 2010): $C_{PURCHASE} = 28,000 + 54.0 A^{1.2}$ $A [=] m^2$

Heat Transfer Area: A

Lang Factor: $LF = 3.5$

Annualized Capital Charge Ratio: $ACCR = 0.2$

Annualized Capital Cost (\$/year): $ACC = C_{PURCHASE} * LF * ACCR$

Cost of Cooling Water: $C_{CW} = \$14.8 / 1000 m^3$

Overall Heat Transfer Coefficient: $U = 425.6 W / (m^2 \cdot K)$

Heat Capacity of Water: $C_p = 4.184 kJ / (kg \cdot K)$

Operating Year: 8000 hrs

Cost Escalation Factor: 1.50

Section #1: PFD

The single-pass condenser was modelled using a FAKE heat exchanger, which revolves around using two heater units in parallel, rather than a single heat exchanger. This was done mainly to simplify the design specifications of the unit. A hierarchy using a real heat exchanger was used to import the heat exchanger area when calculating capital costs. Both the diagram for the FAKE and REAL exchanger are shown in figure 1 and figure 2, respectively.

Figure 1: Main Process Flow Diagram

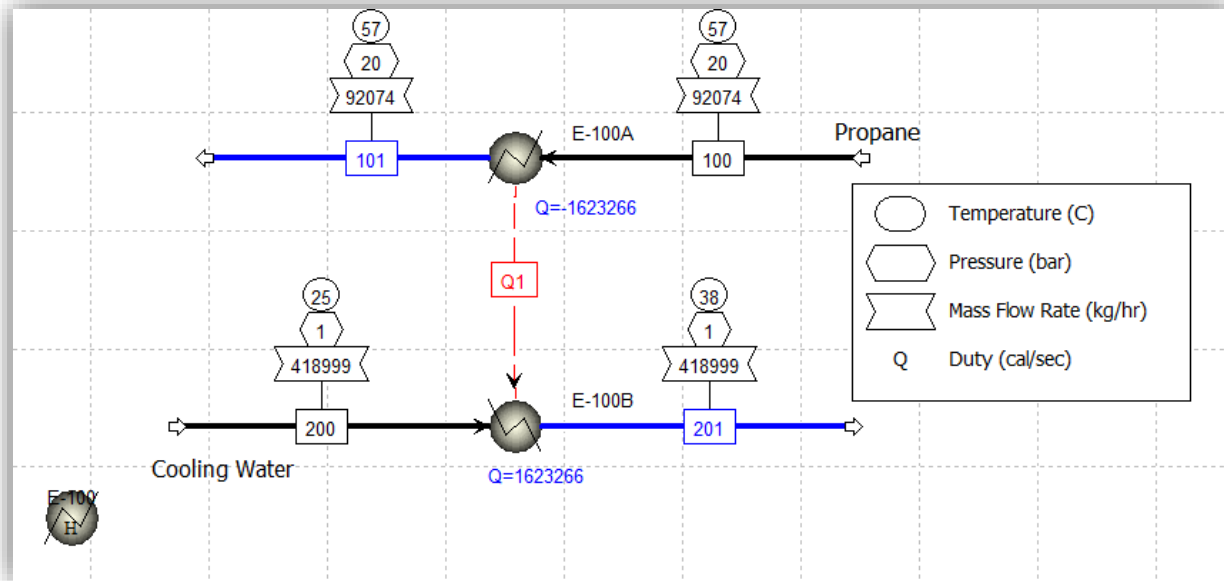
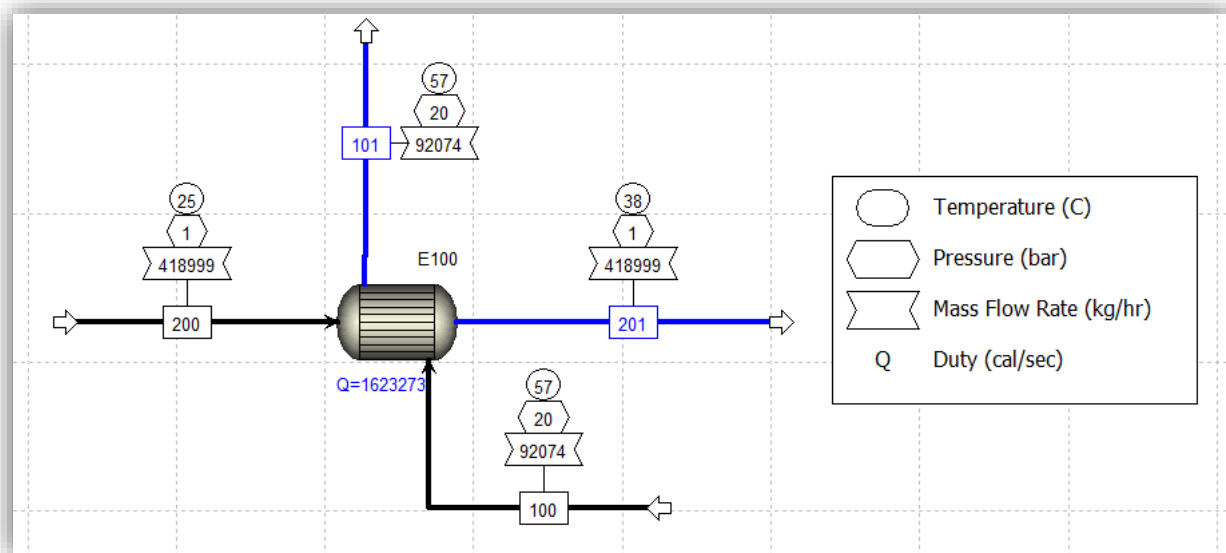


Figure 2: REAL HEX Process Flow Diagram



Section #2: Heat and Weight Balance

Table 1 presents the stream summary output from Aspen Plus V14 for the condensation system. The data were exported to Excel and reformatted for clarity and consistency with the UIC template. Four unused rows (Description, MIXED Substream, Mass Fractions, and Mole Fractions) were removed since they contained no data.

The system was modeled using two heaters: one for the propane stream (Streams 100–101) and one for the water stream (Streams 200–201). A heat exchanger was later included in a hierarchy for cost estimation, which produced similar results as the two-heater configuration.

Table 1: Stream Summary

Stream Property	Units	100	101	200	201
Phase		Vapor Phase	Liquid Phase	Liquid Phase	Liquid Phase
Temperature	C	57.0047	56.8406	25.0000	37.9116
Pressure	bar	20.0000	19.9311	1.0000	0.9311
Molar Vapor Fraction		1.0000	0.0000	0.0000	0.0000
Mass Flows	kg/sec	25.5760	25.5760	116.3886	116.3886
Mole Flows	kmol/hr	2088.0000	2088.0000	23257.9813	23257.9813
Molar Enthalpy	cal/mol	-25024.6376	-27823.3720	-68725.8549	-68474.5967
Molar Entropy	cal/mol-K	-69.6776	-78.1542	-40.1199	-39.2948
Enthalpy Flow	cal/sec	-14514289.7859	-16137555.7676	-444006845.6562	-442383579.6745
Average MW		44.0965	44.0965	18.0153	18.0153
Molar Density	mol/cc	0.0011	0.0099	0.0552	0.0545
Heat capacity, mixture	cal/mol-K	25.0472	37.8253	19.4663	19.4619
Availability (flow), mixture	cal/sec	-2465143.3381	-2622585.1611	-366727455.8248	-366693512.7795
Exergy flow rate	cal/sec	919350.9473	761909.1244	-43.2960	33899.7483

Section #3: Solution Logic and description

The solution logic for this condensation system was developed using Aspen Plus V14. The process was modeled with two heaters representing the hot and cold sides of the heat exchanger, and a separate exchanger was later introduced within a hierarchy solely for cost estimation purposes. A calculator block was used to link the heat exchanger area and water mass flow rate to a Fortran-based cost model, while an optimizer was implemented to minimize the total cost of operation.

The flowsheeting structure consists of two heaters arranged in parallel. The propane heater represents the hot side, where the inlet stream was specified at 20 bar and a vapor fraction of 1.0 to simulate propane entering at its dew point. The outlet vapor fraction was fixed at 0.0 to condense the propane completely, and a small pressure drop is simulated by setting the pressure to -1 psi. The water heater represents the cold side, with an inlet temperature of 25 °C and a pressure of 1 bar. A similar pressure drop was applied across this heater. The heat stream travels from the hot side (propane) to the cold side (water). These two heaters simulate the thermal behavior of the real exchanger, while the exchanger block in the hierarchy mirrors their duties and is used to extract the heat transfer area for cost analysis.

Two Transfer blocks (one for the hot side and one for the cold side) copy the relevant inlet stream states from the two-heater model to the inlet streams of the real heat exchanger in the hierarchy, ensuring the exchanger is sized under the same conditions. In that exchanger, a pressure drop is set, the hot-outlet vapor fraction is fixed at 0.0 and the overall U is specified; the area computed there is exported.

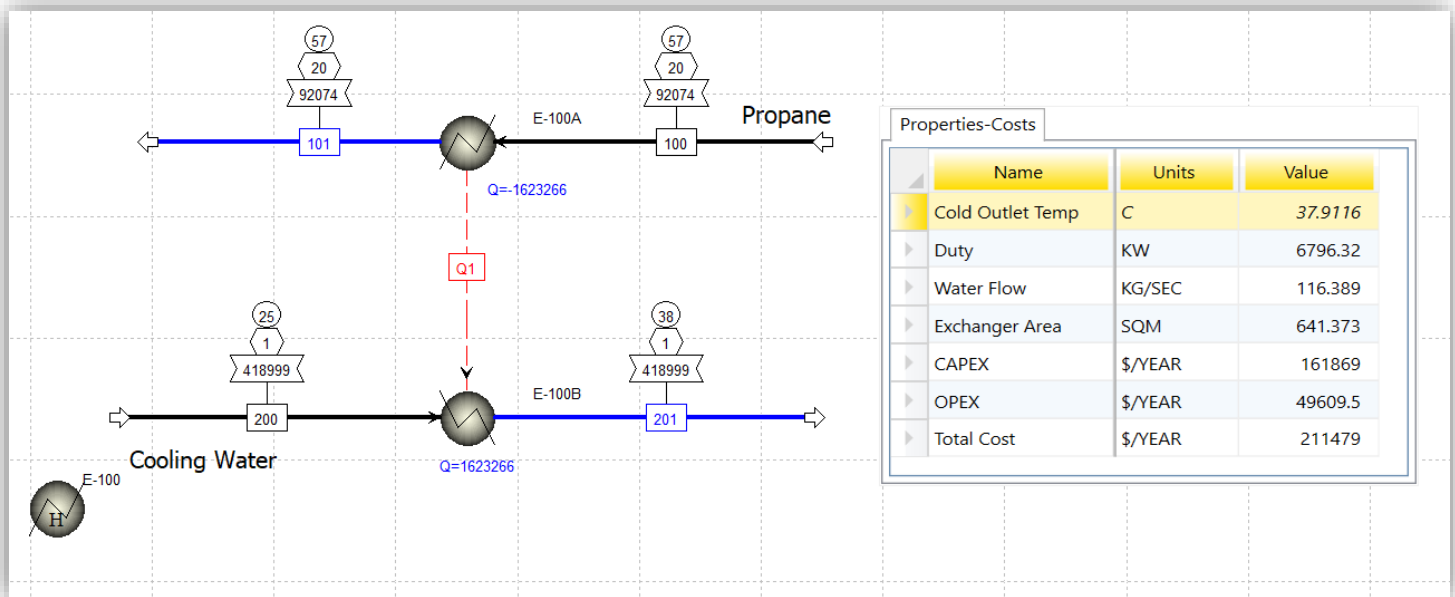
A calculator block was implemented to compute the capital, operating, and total costs of the heat exchanger. The purchase cost was determined using the provided purchase cost equation as a function of area, and the capital cost was then obtained by applying the Lang factor, annual capital charge rate (ACCR), and cost escalation factor. The operating cost was derived by calculating the annual cooling water volume from the water mass flow rate and density, then multiplying by the unit cost of \$14.8 per 1000 m³. The total cost was found by summing the CAPEX and OPEX. Among the imported variables, only the water mass flow rate and heat exchanger area were directly used in the Fortran code, while other variables (such as duty) were transferred for result display.

An optimization block was used to minimize the total cost by varying the inlet cooling water mass flow rate. A constraint was applied to keep the outlet temperature of the cooling water below 45 °C, ensuring that the actual temperature rise remained within the expected range of approximately 15 °C. It was observed that the optimizer's final result varied slightly (by 1–2%) depending on the initial mass flow rate guess specified in the inlet stream; guesses closer to the optimal value yielded results more quickly and with higher accuracy.

Section #4: Miscellaneous results

While no specific KPIs were required, several key thermodynamic and cost parameters were compiled into a custom table for clarity and documentation. After executing the optimization routine, the optimal cooling-water circulation rate was found to be approximately **116 kg/s**, corresponding to a temperature rise of about 13 °C across the exchanger. The results show that the capital cost (\$161,869/year) is significantly higher than the operating cost (\$49,609/year), which is expected since the exchanger's large surface area and construction dominate initial expenses, while utility costs remain relatively low. The combined total annual cost was calculated to be \$211,479.

Figure 3: Significant Properties and Costs custom table



Section 5: References

Note that since no additional discussion was requested for this assignment, the references section replaces the discussion section.

This problem was relatively straightforward, and all information used to solve it was derived from the materials provided in the problem description or from lessons taught by Dr. Michael Caracotsios. No external sources or research articles were consulted, however, the cost of equipment used information from Gavin's textbook. Aspen Plus V14 was used to perform the simulation and costing calculations, while ChatGPT was utilized to assist with navigating Aspen and refining the clarity and formatting of this report.

References

Aspen Technology, Inc. (2024). *Aspen Plus V14* [Computer software]. AspenTech.
<https://www.aspentech.com/>

OpenAI. (2025). *ChatGPT (GPT-5)* [Large language model]. OpenAI. <https://chat.openai.com/>

Towler, Gavin, and Ray Sinnott. *Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design*. 3rd ed. Butterworth-Heinemann, 2021.

Problem B: Optimization of a refrigeration loop

Methane at temperature 25 °C and pressure 30 bar is to be precooled down to -38 °C prior to liquefaction. The precooling will be done using propane as the refrigerant. Design a multi-stage refrigeration loop and calculate the annual operating cost and the capital cost. You may attempt to minimize the total annualized cost for your design. The flow rate of methane is 250 MMSCFD. The isentropic efficiency of the compressors can be taken as 80% and the mechanical efficiency equal to 95%.

Notes:

For both problems assume that the physical properties of methane and propane are best described by the Peng-Robinson Equation of State

Minimum temperature approach for condenser may be taken as $\Delta T_{\min} = 10^\circ\text{C}$

Minimum temperature approach for evaporators:

Stage	$\Delta T_{\min}$
1	8
2	6
3	4
4	2

Gavin Towler and Ray Sinnott, **Chemical Engineering Design**. Principles Practice and Economics of Plant and Process Design, Second Edition © Elsevier Ltd. 2013.

Data:

Heat Exchangers: Shell and Tube Single Pass

Purchase cost of the Heat Exchanger (\$ January 2010): $C_{\text{PURCHASE}} = 28,000 + 54 A^{1.2}$ $A [=] m^2$

Heat Transfer Area: A

Lang Factor: $LF = 3.5$

Compressors: Centrifugal

Purchase cost of the Compressor (\$ January 2010): $C_{\text{PURCHASE}} = 580,000 + 20,000 P^{0.6}$ $P [=] kW$

Power Requirement: P

Lang Factor: $LF = 2.5$

Valves: Expansion Valves

Purchase cost of the Valve (\$ January 2010): $C_{\text{PURCHASE}} = 8,142 m^{0.7}$ $m [=] kg / s$

Mass Flow Rate: m

Lang Factor: $LF = 1.3$

Drums: Flash Drum with Hemispherical Head and Bottom

Purchase cost of the Drum (\$ January 2010): $C_{PURCHASE} = 11,600 + 34W^{0.85}$ $W [=] kg$

Shell Mass: W

Lang Factor: $LF = 4.0$

Data (Continued):

Annualized Capital Charge Ratio: $ACCR = 0.2$

Annualized Capital Cost (\$/year): $ACC = C_{PURCHASE} * LF * ACCR$

Cost of Cooling Water: $C_{CW} = \$0.354 / GJ$

Cost of Electricity: $C_{EL} = \$0.06 / kWh$

Overall Heat Transfer Coefficient Condenser: $U = 425.6 W / (m^2 \cdot K)$

Overall Heat Transfer Coefficient Evaporators: $U = 425.6 W / (m^2 \cdot K)$

Heat Capacity of Water: $C_p = 4.184 kJ / (kg \cdot K)$

Cooling Water Temperature: $T_{CWin} = 25 ^\circ C$

Cooling Water Pressure: $P_{CWin} = 20 psia$

Cooling Water Temperature Rise: $10 ^\circ C$

Operating Year: $8000 hrs$

Cost Escalation Factor: 1.50

Additional Instructions for your Report:

Your Custom table should contain three tabs

Tab 1 (Name KPIs): Display COP, Total Power(kW), Total Annualized Capital Cost (MM\$/oper_year), Total Utility Cost (MM\$/oper_year) and the sum of the two costs

Tab2 (Name: Detailed Cost): The installed cost (\$) of each equipment and cost of each utility in \$/oper_year

Tab3 (Name: Optimal Pressure): The final discharge pressure of each stage in bar

Section 1: PFD

Out of all the different compressor stages for the refrigerant cycle, we found that a 4-stage compressor is the most cost-efficient. The cost was determined using the constants and equations provided in this assignment, with the fake units and the hierarchy set up to match the real heat exchanger unit.

Figure 1: Process Flow Diagram

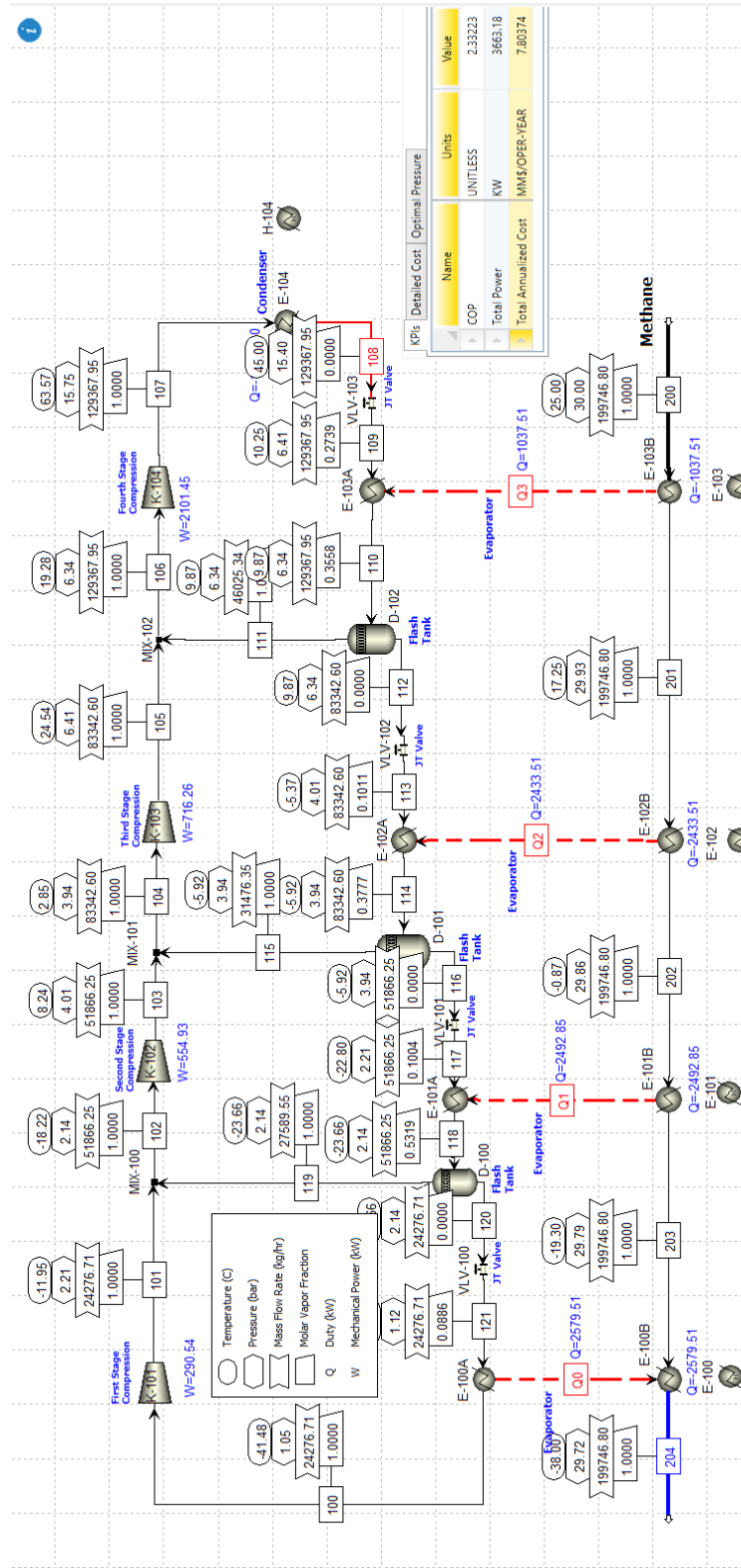


Figure 2: Evaporator Hierarchy

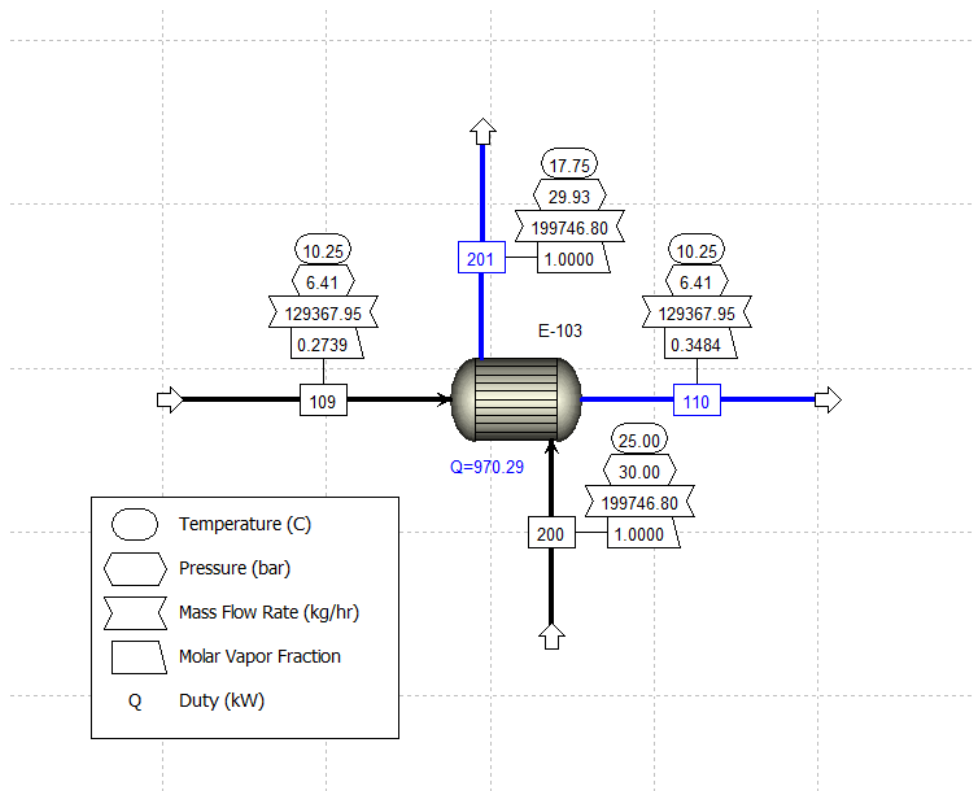
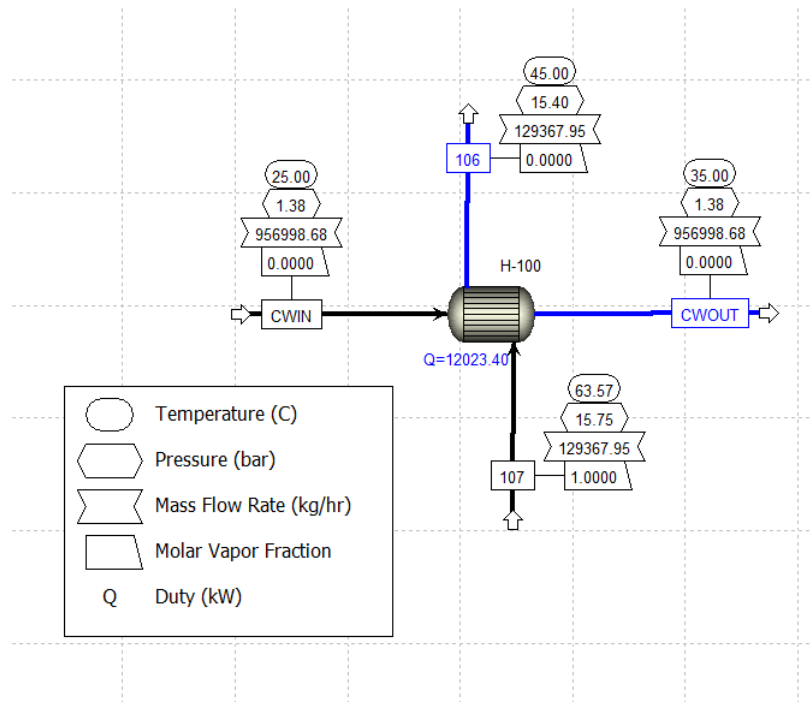


Figure 3: Condenser Hierarchy



Section #2: Heat and Weight Balance

The UIC template was used to display the stream results, and images were taken directly from Aspen Plus V14 due to the extensive amount of data, which make it hard to format it properly.

Table 1: Stream summary (100-108)

	Units	100	101	102	103	104	105	106	107	108
Description										
- MIXED Substream										
Phase										
Temperature	C	-41.4765	-11.9491	-18.2198	8.23688	2.84991	24.5385	19.2781	63.569	44.9998
Pressure	bar	1.04748	2.20892	2.13997	4.00886	3.93991	6.40875	6.3398	15.748	15.4033
Molar Vapor Fraction		1	1	1	1	1	1	1	1	0
+ Mass Flows	kg/hr	24276.7	24276.7	51866.3	51866.3	83342.6	83342.6	129368	129368	129368
+ Mass Fractions										
+ Mole Flows	kmol/hr	550.536	550.536	1176.2	1176.2	1890.01	1890.01	2933.78	2933.78	2933.78
+ Mole Fractions										
Molar Enthalpy	kJ/kmol	-109401	-107596	-108019	-106406	-106792	-105495	-105895	-103446	-118199
Molar Entropy	kJ/kmol-K	-287.242	-285.846	-287.236	-286.081	-287.333	-286.457	-287.73	-286.263	-332.347
Enthalpy Flow	kW	-16730.3	-16454.3	-35292.2	-34765.1	-56065.8	-55385.3	-86298.3	-84301.9	-96325.3
Average MW		44.0965	44.0965	44.0965	44.0965	44.0965	44.0965	44.096	44.096	44.096
Molar Density	kmol/cum	0.0563098	0.107484	0.1069	0.186566	0.187575	0.292415	0.296151	0.713867	10.3913
Heat capacity, mixture	kJ/kmol-K	65.1222	69.5395	68.7203	73.8317	73.0756	78.2861	77.5869	93.1035	141.65
Availability (flow), mixture	kW	-3633.5	-3421.14	-7311.93	-6897.35	-11089.8	-10546.4	-16387.4	-14747.4	-15573.7
Energy flow rate	kW	107.779	319751	681178	1095.06	1754.29	2296.83	3551.32	518916	4432.31

Table32: Stream summary (109-116)

	Units	109	110	111	112	113	114	115	116	117
Description										
→ MIXED Substream										
Phase				Vapor Phase	Liquid Phase			Vapor Phase	Liquid Phase	
Temperature	C	10.2522	9.87124	9.87124	9.87124	-5.37282	-5.91604	-5.91604	-5.91604	-22.8028
Pressure	bar	6.40875	6.3398	6.3398	6.3398	4.00886	3.93991	3.93991	3.93991	2.20892
Molar Vapor Fraction		0.273874	0.355778	1	0	0.101082	0.377675	1	0	0.100399
+ Mass Flows	kg/hr	129368	129368	46025.3	83342.6	83342.6	83342.6	31476.4	51866.3	51866.3
+ Mass Fractions										
+ Mole Flows	kmol/hr	2933.78	2933.78	1043.77	1890.01	1890.01	1890.01	713.808	1176.2	1176.2
+ Mole Fractions										
Molar Enthalpy	kJ/kmol	-118199	-116926	-106619	-122618	-122618	-117983	-107427	-124389	-124389
Molar Entropy	kJ/kmol-K	-331.185	-326.663	-290.242	-346.778	-346.534	-329.173	-289.673	-353.145	-352.861
Enthalpy Flow	kW	-96325.3	-95287.8	-30912.9	-64374.8	-64374.8	-61941.3	-21300.7	-40640.6	-40640.6
Average MW		44.096	44.096	44.0951	44.0965	44.0965	44.0965	44.0964	44.0965	44.0965
Molar Density	kmol/cum	1.06956	0.832594	0.310475	11.6811	1.717	0.504318	0.195508	12.1765	1.04171
Heat capacity, mixture	kJ/kmol-K	105.096	101.729	76.6022	115.606	105.24	94.8889	72.0497	108.749	99.6189
Availability (flow), mixture	kW	-15856	-15917	-5823.09	-10093.9	-10132.2	-10416.1	-4176.06	-6240.01	-6267.63
Exergy flow rate	kW	4148.23	4080.39	1272.46	2807.9	2769.45	2468.67	675.736	1792.93	1765.15

Table 3: Stream summary (118-204)

	Units	118	119	120	121	200	201	202	203	204
Description										
- MIXED Substream										
Phase			Vapor Phase	Liquid Phase		Vapor Phase	Vapor Phase	Vapor Phase	Vapor Phase	Vapor Phase
Temperature	C	-23.6629	-23.6629	-23.6629	-40	25	17.2522	-0.872818	-19.3028	-38
Pressure	bar	2.13997	2.13997	2.13997	1.11643	30	29.9311	29.8621	29.7932	29.7242
Molar Vapor Fraction		0.531936	1	0	0.0886302	1	1	1	1	1
+ Mass Flows	kg/hr	51866.3	27589.5	24276.7	24276.7	199747	199747	199747	199747	199747
+ Mass Fractions										
+ Mole Flows	kmol/hr	1176.2	625.663	550.536	550.536	12450.9	12450.9	12450.9	12450.9	12450.9
+ Mole Fractions										
+ Mole Fractions										
Molar Enthalpy	kJ/kmol	-116759	-108392	-126269	-126269	-75066.3	-75366.3	-76069.9	-76790.7	-77536.5
Molar Entropy	kJ/kmol-K	-322.252	-288.713	-360.368	-360.099	-110.059	-111.061	-113.545	-116.269	-119.304
Enthalpy Flow	kW	-38147.8	-18837.9	-19309.8	-19309.8	-259623	-260661	-263094	-265587	-268166
Average MW		44.0965	44.0965	44.0965	44.0965	16.0428	16.0428	16.0428	16.0428	16.0428
Molar Density	kmol/cum	0.204537	0.109628	12.6932	0.643954	1.29171	1.33151	1.44285	1.5827	1.76547
Heat capacity, mixture	kJ/kmol-K	84.3803	68.113	102.867	95.7664	38.9101	38.8669	38.9821	39.49	40.6264
Availability (flow), mixture	kW	-6756.52	-3877.69	-2878.83	-2891.1	-146133	-146138	-146009	-145694	-145143
Exergy flow rate	kW	1257.75	374.608	883.142	870.797	28699.2	28700.9	28844.9	29178.3	29748.4
<add properties>										

Table 4: Condenser Hierarchy stream summary

	Units	106	107	CWIN	CWOUT	
Description						
- MIXED Substream						
Phase		Liquid Phase	Vapor Phase	Liquid Phase	Liquid Phase	
Temperature	C	44.9998	63.569	25	34.9997	
Pressure	bar	15.4033	15.748	1.37895	1.37895	
Molar Vapor Fraction		0	1	0	0	
+ Mass Flows	kg/hr	129368	129368	956999	956999	
+ Mass Fractions						
+ Mole Flows	kmol/hr	2933.78	2933.78	53121.5	53121.5	
+ Mole Fractions						
Molar Enthalpy	kJ/kmol	-118199	-103446	-287741	-286926	
Molar Entropy	kJ/kmol-K	-332.347	-286.263	-167.974	-165.286	
Enthalpy Flow	kW	-96325.3	-84301.9	-4.24589e+06	-4.23387e+06	
Average MW		44.096	44.096	18.0153	18.0153	
Molar Density	kmol/cum	10.3913	0.713867	55.173	54.6346	
Heat capacity, mixture	kJ/kmol-K	141.65	93.1035	81.5008	81.4772	
Availability (flow), mixture	kW	-15573.7	-14747.4	-3.50689e+06	-3.5067e+06	
Exergy flow rate	kW	4432.31	5189.16	18.3285	142.171	
<add properties>						

Table 5: Heat Stream summary

	Q0	Q1	Q2	Q3	
QCALC kW	-2579.50514	2492.85315	2433.51222	1037.50657	
TBEGIN C	-41.4764714	-0.872817543	17.2521696	25	
TEND C	-40.000004	-19.3028476	-0.872817543	17.2521696	

Tables 1 through 3 show the stream summaries in the main flowsheet, and Table 4 displays the hierarchy stream summary for the condenser, where we used cooling water and fed it as a stream rather than a utility. The other hierarchies, like the evaporators, are not explicitly included, as they contain the same results in the main flowsheet. Table 5 displays the heat summary of the heat transfer streams used for the evaporators.

Section #3: Solution Logic and Description

The refrigeration loop is a four-stage propane cycle that precools/condenses methane across four evaporators (E-100 A/B, E-101 A/B, E-102 A/B, E-103 A/B) with interstage flash drums and throttling valves, followed by a condenser. Compression is in four stages with intercooling/mixing, and a condenser removes the total heat load before recycle.

The Aspen model uses explicit unit models on the main flowsheet and five hierarchies for exchanger sizing/costing: one hierarchy per evaporator (Stages 1–4) and one for the condenser. In each evaporator hierarchy the exchanger spec is (hot outlet – cold inlet) temperature difference; in the condenser hierarchy the spec is hot-side outlet vapor fraction = 0 (complete condensation). The inlets to the parallel heaters (evaporators) are transferred to their respective hierarchies via Transfer blocks so the hierarchy exchangers size under the same conditions, and the propane inlet to the condenser is likewise transferred to the condenser hierarchy; cooling-water inlet data are entered per the assignment. Because a CW mass-flow was not provided, a Design Spec sets CW outlet = 35 °C (25 °C inlet with 10 °C rise) by adjusting CW inlet mass-flow. The last valve pressure (VLV-100, leftmost) is fixed from a purity-based saturation calculation of propane at –40 °C (matching the Stage-4 evaporator at –38 °C with $\Delta T_{\min} = 2$ °C). A second Design Spec enforces methane outlet = 38 °C by adjusting the propane flow in the tear stream. Thus, there are two design specs and five exchanger hierarchies tied to the main flowsheet via Transfers.

Calculator Blocks

1. C-RES — performance metrics

This calculator imports the mechanical power of each compressor (W_i) and the duty of each evaporator ($Q_{e,j}$), and computes:

$$W_{\text{tot}} = \sum_i W_i, \quad Q_{\text{evap}} = \sum_j Q_{e,j}, \quad \text{COP} = \frac{Q_{\text{evap}}}{W_{\text{tot}}}.$$

It exports COP, total power, and any intermediate sums for reporting.

2. C-SIM — $\Delta T_{\min}$ and pressure propagation

C-SIM applies the $\Delta T_{\min}$ constraints and synchronizes pressures. It imports the cold-inlet temperatures to the first three evaporators (propane inlet to the top heater in each pair) and sets the hot-outlet temperature of the corresponding bottom exchangers (methane side) to $T_{\text{cold,in}} + \Delta T_{\min}$. After manual adjusting the $\Delta T_{\min}$ values of each stage, the values Stage 1: 7 °C, Stage 2: 4.5 °C, Stage 3: 3.5 °C yielded lower costs than the problem's suggested values; these same ΔT targets are also manually set inside the evaporator hierarchies' "hot-outlet – cold-inlet temperature difference" specs. C-SIM also exports imported valve discharge pressures to the compressors as for staging consistency (all except the fourth-stage compressor, which is not overwritten, and VLV-100, which does not transfer a pressure).

3. C-COSTS — equipment and utility economics

C-COSTS aggregates installed, capital (annualized), and operating costs by equipment and totals. Inputs:

- Exchanger areas from all five hierarchies → exchanger purchase cost via the provided area-based correlation; then installed/annualized capital (ACC) using the Lang factor, escalation, and ACCR from the problem statement.
- Compressor powers → compressor costs per the given compressor correlation.
- Condenser duty → cooling-water OPEX using the stated unit price (in \$/G/J, then convert).
- Valve inlet mass-flows → valve CAPEX via the valve cost correlation.
- Flash drum weights → drum CAPEX (weights taken from one economic run and assumed constant).
- Electricity OPEX via Aspen utility variable for Cost-Rate.

C-COSTS reports per-equipment Installed and ACC, CW and electricity costs, and totals (Installed, ACC, Total Annualized Cost). (*Installed $\cong$ ACC without the ACCR term*)

Optimizer

An Optimization block was used to minimize total compressor work. Decision variables were:

1. Valve pressures for VLV-101, VLV-102, VLV-103 (the three valves that transfer pressure), varied within reasonable bounds such that pressures decrease from right→left across valves and increase through compressor stages;
2. Compressor pressure ratios for Stages 1–3 (compressors modeled in terms of pressure ratio rather than absolute pressure).

All other constraints from the design specs (e.g., CW outlet 35 °C; methane outlet 38 °C) and $\Delta T_{\min}$ targets remain active, ensuring feasible temperature approaches and stage matching while the optimizer searches for the minimum W_{tot} . Note that running this optimizer while only varying the pressure of the valves without varying pressure ratios results in a much higher price.

Section #4: Miscellaneous results

Figure 3: Key Performance Indicators

KPIs	Detailed Cost	Optimal Pressure	
	Name	Units	Value
▶	COP	UNITLESS	2.33223
▶	Total Power	KW	3663.18
▶	Total Annualized Cost	MM\$/OPER-YEAR	7.80374

Figure 3 displays the custom table of key performance indicators (KPIs) where our optimized COP (coefficient of performance) is 2.33233 our total power is 3663.18 KW, and the total annualized cost is 7.80374 MM\$/operating year.

Figure 4: Detailed Cost

KPIs	Detailed Cost	Optimal Pressure	
	Name	Units	Value
▶	Compressors (4)	\$/OPER-Y...	9.56378e+06
▶	Heat Exchnagers (...)	\$/OPER-Y...	501167
▶	Valves (4)	\$/OPER-Y...	5.49855e+06
▶	Drums (3)	\$/OPER-Y...	6.04851e+06
▶	Total Installed Cost	\$/OPER-Y...	303494
▶	Electricity	\$/OPER-Y...	157131
▶	Cooling Water	\$/OPER-Y...	520284

Figure 4 displays the installed cost of the 4 compressors, 5 heat exchangers, 4 valves, 3 drums, and the Total installed cost, as well as the cost of each utility.

Figure 5: Optimal Pressures

KPIs	Detailed Cost	Optimal Pressure

Figure 5 Displays the optimal discharge pressure of each compressor, which showed a pressure ratio of about 1.9. Note that there might be slight differences in pressure ratio between compressors, since the system was directly modelled in terms of pressure ratio.

Section 5: References

All information used to solve this problem was derived from the materials provided in the problem description or from lessons taught by Dr. Michael Caracotsios. No external sources or research articles were consulted, however, the cost of equipment used information from Gavin's textbook. Aspen Plus V14 was used to perform the simulation and costing calculations, while ChatGPT was utilized to assist with navigating Aspen and refining the clarity and formatting of this report.

References

Aspen Technology, Inc. (2024). *Aspen Plus V14* [Computer software]. AspenTech.
<https://www.aspentech.com/>

OpenAI. (2025). *ChatGPT (GPT-5)* [Large language model]. OpenAI. <https://chat.openai.com/>

Towler, Gavin, and Ray Sinnott. *Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design*. 3rd ed. Butterworth-Heinemann, 2021.

Problem 1: Biomass Pyrolysis Simulation

Biomass conversion processes that are based on pyrolysis offer high yields of liquid transportation fuels such as gasoline and diesel blend stocks, or refinery intermediates. Fast pyrolysis proceeds by heating the biomass to an intermediate temperature where volatilization begins but the gasification of vapor-phase products does not occur. Fast pyrolysis generates a liquid bio-oil that can be transported efficiently and upgraded to fuels.

In a conceptual process design (see Process Flowsheet Diagram Figure 1. below), the fast pyrolysis reactor system is usually represented as a dual circulating fluidized bed. Biomass and hot sand are introduced into the bottom of a vertical riser reactor. As the solids are carried up the riser by a fluidizing gas, the rapid heating of biomass particles to around 450-500 degrees Celsius, produces non-condensable gases and vapors of oxygenated hydrocarbons. Unreacted biomass flows out as solid char. Solid and vapor phases exit the top of the riser and are separated in a cyclone. Sand and char drop into a char combustor, which is basically a second fluidized bed where the unreacted char is combusted in air to reheat the sand. The energy to drive the endothermic pyrolysis reactions is thus provided via the hot sand, by partial combustion of the biomass itself. The vapor phase is condensed downstream to form the bio-oil. In some cases, the bio-oil separates into organic and aqueous phases. Fast pyrolysis bio-oils can be as high as 50% oxygen by mass and are generally assumed to be hydro-treated downstream, to reduce oxygen.

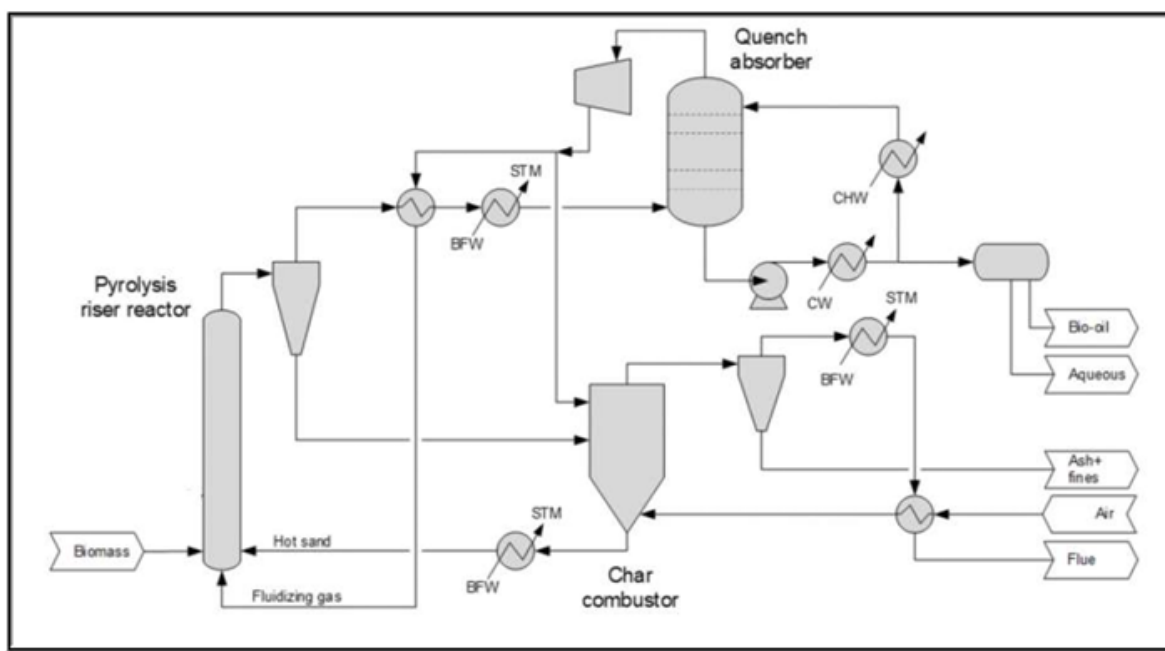


Figure 1: Process Flow Diagram(PFD) of Biomass Fast Pyrolysis

The objective of the assignment is to prepare an ASPEN Plus simulation and calculate:

1. Total Power(KW) consumed
2. Total High(250 °C) and Low(125 °C) pressure steam generated (MMSCFD)
3. Mass yield (%) of incoming carbon to **Bio-oil**
4. Mass yield (%) of incoming carbon to **Flue Gas**
5. Mass yield (%) of incoming carbon to **Waste Water**

Your final report should also include the following

1. Material and Energy Balance with the UIC template
2. A brief outline of your methodology for modeling the pyrolysis reactor
3. A brief outline of your methodology for modeling the char combustor

The Quench Absorber can be simulated using a SEP2 Aspen Plus Unit Operation block

Data for the simulation

1. Bone dry biomass flow rate: **50,000 kg/hr**
2. Biomass feed temperature: **25 deg C**
3. Biomass feed pressure: **2.3 atm**
4. Moisture weight fraction: **0.1**
5. Fluidization gas to Sand mass ratio: **0.1**
6. Bleed Gas to Combustor: **13,000 kg/hr**
7. Bleed Gas temperature: **80 deg C**
8. Bleed Gas pressure: **2.31 atm**
9. Sand to dry biomass mass ratio: **7.0**
10. Biomass **PROXANAL**
 - a. Moisture:
 - b. Fixed Carbon (FC): $0.85 \cdot (100 - \text{ASH})$
 - c. Volatile Matter (VM): $0.15 \cdot (100 - \text{ASH})$
 - d. Ash (ASH): 0.3000
11. Biomass **ULTANAL**
 - e. ASH: 0.30
 - f. CARBON: 49.66
 - g. HYDROGEN: 6.31
 - h. NITROGEN: 0.10
 - i. CHLORINE: 0.00
 - j. SULFUR: 0.08
 - k. OXYGEN: 43.55
12. Biomass **SULFANAL**
 - l. PYRITIC: 0.00
 - m. SULFATE: 0.04
 - n. ORGANIC: 0.04

13. Pyrolysis Reactor Pressure Drop:	0.3 atm
14. Pyrolysis Reactor Temperature:	480 deg C
15. Pyrolysis Reactor Yield (from Experiments)	
o. Gas:	12.9%
i. H ₂ :	0.10%
ii. CO:	5.82%
iii. CO ₂ :	6.03%
iv. CH ₄ :	0.53%
v. C ₂ H ₄ :	0.42%
p. Water:	14.8%
q. Char:	9.7%
r. Organics:	62.6%
i. Acids[Acetic Acid]:	11.1%
ii. Carbonyls[Acetone]:	12.7%
iii. Phenols[m-Cresol]:	4.0%
iv. Methoxyphenols[Guaiacol]:	27.5%
v. Levoglucosan:	4.4%
vi. Furfural:	2.9%

16. Combustor Pressure: **1.12 atm**

17. Combustion products: **NO₂, SO₂, CO₂, H₂O**

18. Oxygen in the combustor effluent: **2% mol**

19. Fluidization Gas temperature: **375 deg C**

20. Fluidization Gas pressure: **2.31 atm**

21. Fluidization Gas and Bleed Gas composition, **mol%**

s. CO:	31.19
t. CO ₂ :	20.11
u. Water:	15.09
v. NO ₂ :	0.30
w. SO ₂ :	0.91
x. Methane:	2.62
y. Ethylene:	1.51
z. Acetone:	28.27

22. Sand component: SiO₂

23. Temperature of the Quench Absorber (SEP2): 45 deg C

24. Pressure of the Quench Absorber: 1.7 atm

Section #1: PFDs

The full Process flow diagram, which used several hierarchies to hide fake equipment, as well as each hierarchy are shown below.

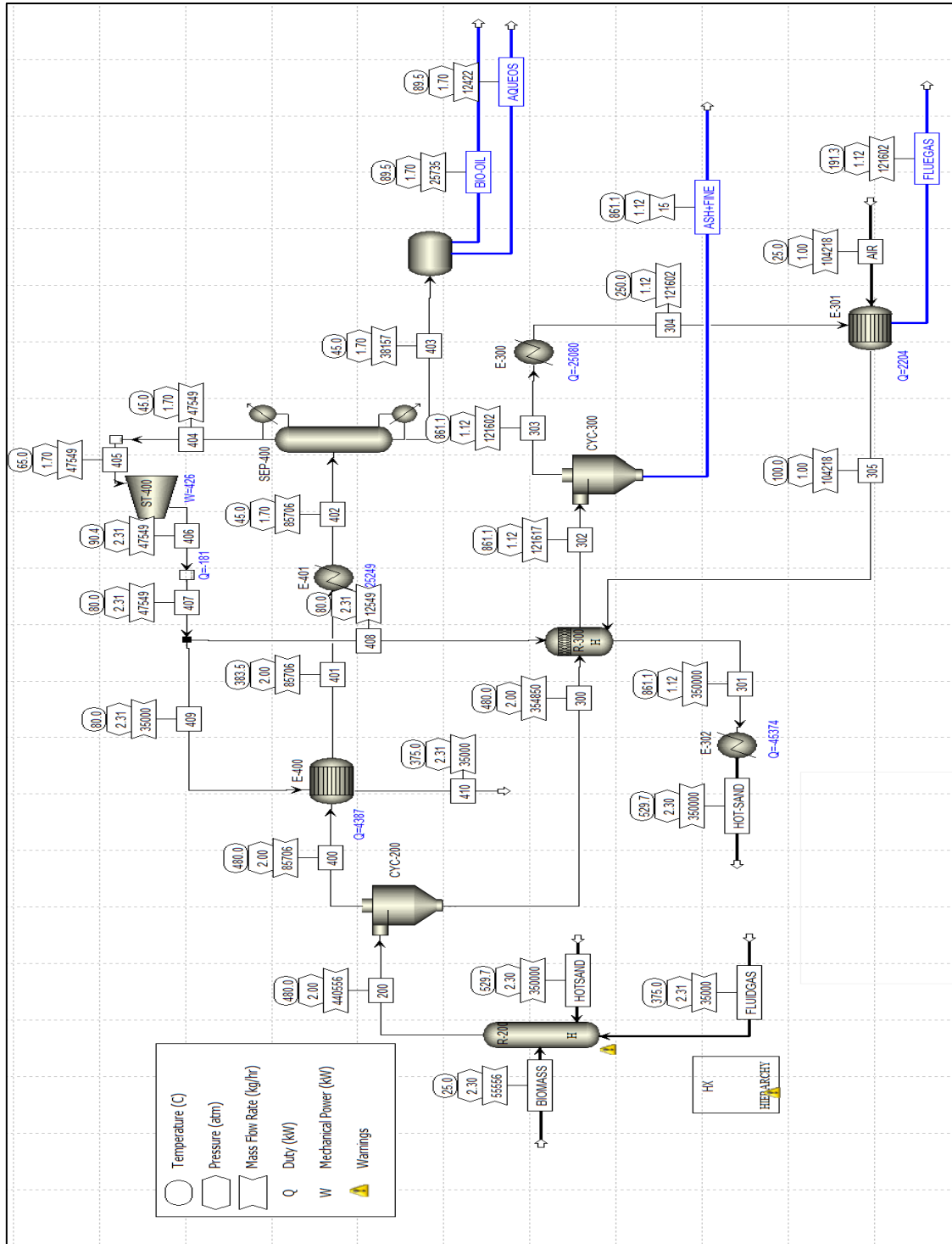
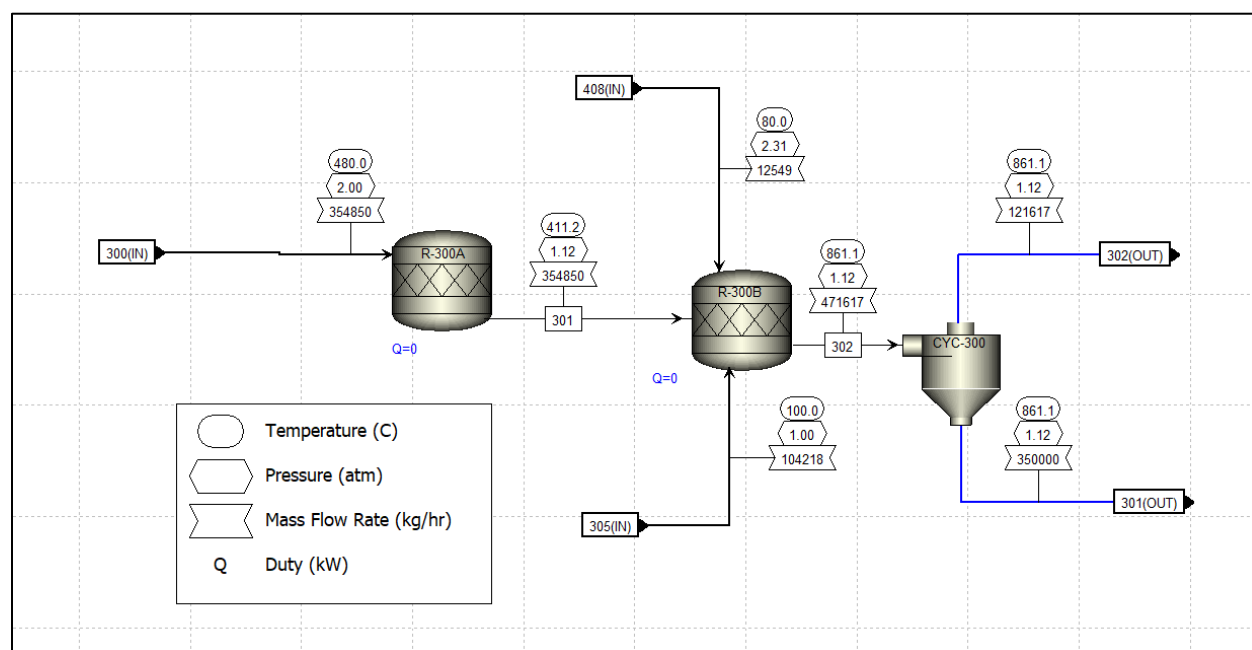
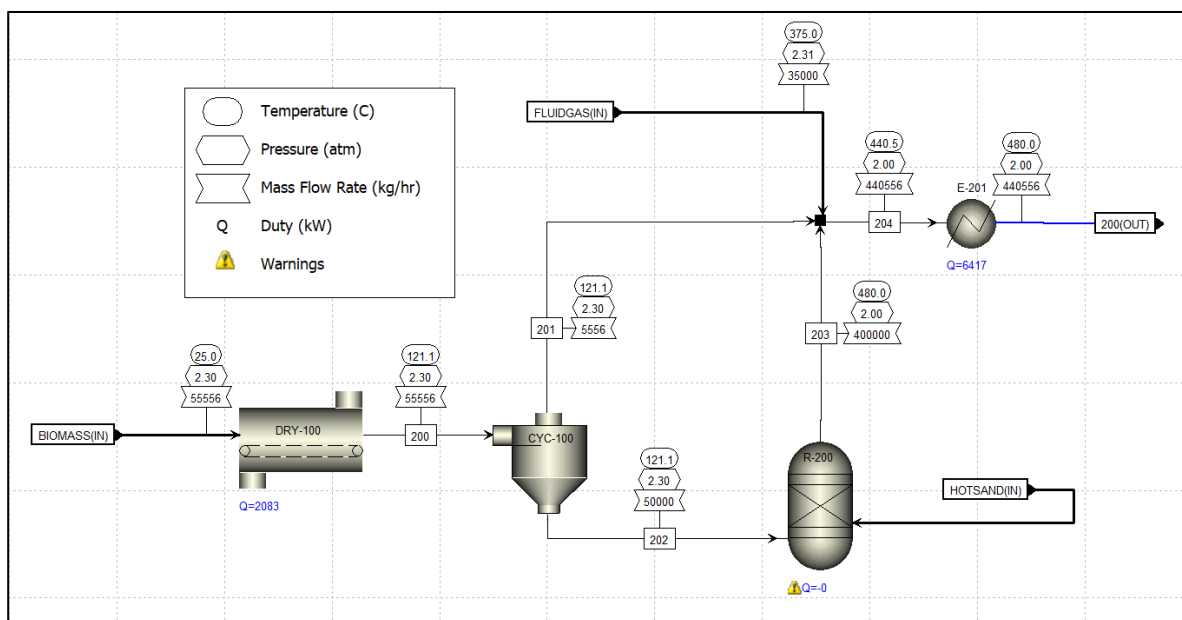
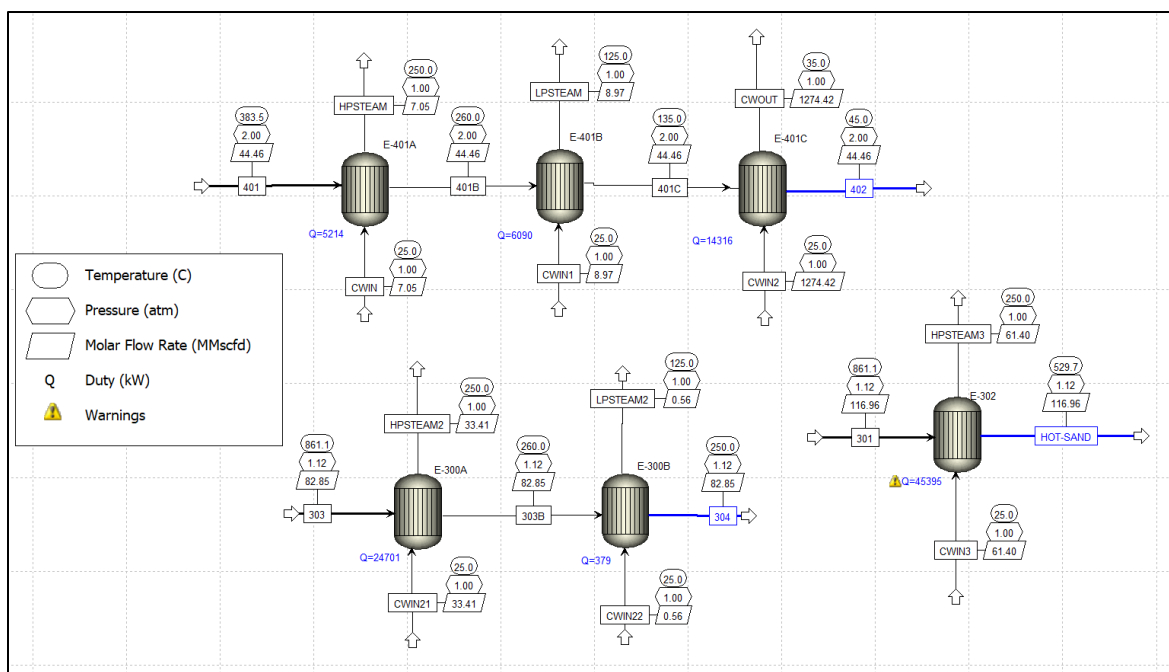


Figure 1: Overall Process Flow Diagram





Section #2: Heat and Weight Balance

The following tables are the stream summaries output from ASPEN V14, in the UIC format.

UIC [Dropdown] | Save | Save As New | Discard Change | Template [Dropdown]

General Options | ☒ Total | ☒ Substreams | ☐ Phases | Stream Summary Options | ☒ Mole | ☒ Mass | Volume | Flows | Composition | Property Sets

Select Properties | Display Options | Calculation Options | Report

Send to Excel/ASW | Send to Flowsheet | Copy All

Capital: ____ USD Utilities: ____ USD/Year Energy Savings: ____ MW (☐) Exchangers - Unknown: 8 OK: 0 Risk: 0 CO₂e Scope 1 **32237 kg/hr** Scope 2 ____

Main Flowsheet x BIOMASS (MATERIAL) - Results (Default) x +

Material	Vol.% Curves	Wt. % Curves	Petroleum	Polymers	Solids	Status
Description						
- Total Stream						
Temperature	C	25	375	529.726	480	480
Pressure	atm	2.3	2.31	2.3	2	2
+ Mass Flows	kg/hr	55555.5	35000	350000	440556	354850
+ Mass Fractions						
Enthalpy Flow	kW	-97119.3	-49590.2	-1.42362e+06	-1.56183e+06	-1.43617e+06
Exergy flow rate						
+ MIXED Substream						
+ CISOLID Substream						
+ NC Substream						
<add properties>						

Material	Vol.% Curves	Wt. % Curves	Petroleum	Polymers	Solids	Status			
		Units	400	401	402	403	AQUEOS	BIO-OIL	
	Description								
	— Total Stream								
	Temperature	C	480	383.474	45	45	89.5196	89.5196	
	Pressure	atm	2	2	1.7	1.7	1.7	1.7	
	+ Mass Flows	kg/hr	85705.5	85705.5	85705.5	38156.7	12421.7	25735	
	+ Mass Fractions								
	Enthalpy Flow	kW	-125660	-130047	-155297	-81108	-53845.6	-27262.4	
	Exergy flow rate								
	+ MIXED Substream								
	+ CISOLID Substream								
	+ NC Substream								
	<add properties>								

Table 2: Separation section stream summary (1)

Material	Vol.% Curves	Wt. % Curves	Petroleum	Polymers	Solids	Status			
		Units	404	405	406	407	408	409	410
Description									
- Total Stream									
Temperature	C		45	65	90.3534	80	80	80	375
Pressure	atm		1.7	1.7	2.31	2.31	2.31	2.31	2.31
+ Mass Flows	kg/hr		47548.8	47548.8	47548.8	47548.8	12548.8	35000	35000
+ Mass Fractions									
Enthalpy Flow	kW		-74510.6	-72761.8	-72335.5	-72516.4	-19138.1	-53378.3	-48990.9
Exergy flow rate									
+ MIXED Substream									
+ CISOLID Substream									
+ NC Substream									
<add properties>									

Table 3: Separation section stream summary (2)

Material	Vol.% Curves	Wt. % Curves	Petroleum	Polymers	Solids	Status					
		Units	300	301	302	303	304	305	AIR	FLUEGAS	ASH+FINE
Description											
− Total Stream											
Temperature	C		480	861.126	861.126	861.126	250	100	25	191.324	861.126
Pressure	atm		2	1.12	1.12	1.12	1.12	1	1	1.12	1.12
+ Mass Flows	kg/hr		354850	350000	121617	121602	121602	104218	104218	121602	14.55
+ Mass Fractions											
Enthalpy Flow	KW		-1.43617e+06	-1.37824e+06	-74864.4	-74864.6	-99944.9	2196.19	-8.20171	-102149	0.176349
Exergy flow rate											
+ MIXED Substream											
+ CISOLID Substream											
+ NC Substream											
<add properties>											

Table 3: Combustion section stream summary

Table 4: Pyrolysis Hierarchy stream summary

Table 5: Combustion Hierarchy stream summary

Section 3: Solution Logic and description

The fast pyrolysis process was modeled in Aspen Plus using the flowsheet and specifications provided in the assignment. Because the conceptual PFD simplifies several operations, the full simulation required additional supporting blocks, assumptions, and organizational structure to reflect realistic process behavior while remaining readable. To keep the final flowsheet clear and consistent with the assignment diagram, three hierarchies were created: one hierarchy for the pyrolysis reactor section, one for the char combustor section, and one for the steam-generation section used to calculate the production of high-pressure and low-pressure steam (reported in MMSCFD). These hierarchies allow the detailed modeling work to remain hidden inside organized subsections, while the top-level flowsheet visually matches the intended simplified PFD.

Properties

The PR-BM property method was selected for this simulation. Several adjustments were made in the Components and NC-Props sections to properly model biomass and solid behavior in Aspen. Two additional components were added: Carbon (as a CI solid) and Sulfur (as a conventional component), which are required for the first RStoic reactor where char is decomposed into elemental species (H_2 , O_2 , Sulfur, Ash, N_2 , and solid Carbon). In addition, ASH was defined as a nonconventional (NC) solid, consistent with typical biomass ash characterization.

All NC solids (WOOD, ASH, and CHAR) were assigned HCOALGEN for enthalpy and DCOALIGT for density. Default option codes were used for all NC components except WOOD, for which the first HCOALGEN option code was set to 6 to enable specification of its heat of combustion. Under Parameters → Pure Components, the heat of combustion (HCOMB) for WOOD was set to 20 MJ/kg, matching common biomass heating values and allowing an adiabatic energy balance to be performed later for the hot sand returning from the combustor.

Although the assignment's PFD illustrates closed loops for sand circulation and fluidizing gas, true recycle loops involving solids can produce strong convergence issues in Aspen. To avoid this, the loops were left open and additional auxiliary equipment was added or specifications were made downstream to emulate the intended loop conditions without compromising solution stability.

methodology for modeling the pyrolysis reactor

The pyrolysis reactor section begins by determining the required wet-biomass feed rate, since the reactor must receive 50,000 kg/hr of bone-dry biomass while the assignment

specifies 10 wt% moisture in the incoming feed. This requires back-calculating the wet-biomass flowrate using

$$\dot{m}_{\text{wet}} = \frac{\dot{m}_{\text{dry}}}{1 - 0.10} \approx 55,556 \text{ kg/hr.}$$

In addition to the flow rate and thermodynamic specifications, the PROXANAL, ULTANAL, and SULFANAL analyses for the biomass were entered exactly as provided in the assignment. The moisture value in PROXANAL was set to 10 wt% to represent the wet-biomass feed. All other values were entered directly as provided. For the char, which appears as a nonconventional component in the reactor effluent, Aspen requires corresponding NC-property data; since no information was provided in the assignment, the char was assumed to have the same PROXANAL, ULTANAL, and SULFANAL composition as dry biomass, but with 0% moisture. This assumption is necessary for Aspen to perform energy and property calculations and is reasonable given the limited information supplied.

The inlet conditions for the biomass (25 °C, 2.3 atm) are taken directly from the problem statement. A DRYER block is used to remove the moisture, with a dryer temperature of 250 °F and a “drying on a wet basis” value of 1×10^{-10} , which ensures essentially complete moisture removal without causing Aspen’s numerical issues associated with setting this value to zero. The dryer output contains a dry NC-solid stream (the bone-dry biomass) and a mixed-phase water stream. These are separated in an SSPLIT block, which separates based on sub streams, and sends the dry biomass (NC) to the reactor and the removed moisture (MIXED) to the downstream mixer.

The hot-sand feed is calculated according to the assignment’s sand-to-dry-biomass mass ratio of 7.0, giving

$$\dot{m}_{\text{sand}} = 7 \times 50,000 = 350,000 \text{ kg/hr.}$$

Likewise, the fluidizing-gas flow is determined from the fluidizing-gas-to-sand ratio of 0.1, giving

$$\dot{m}_{\text{fluidizing gas}} = 0.1 \times 350,000 = 35,000 \text{ kg/hr.}$$

These values are used directly as feed specifications.

The pyrolysis reactor is modeled as an RYIELD block, with mass-fraction yields taken exactly from the assignment. These yields are not atom-balanced, which triggers a warning in Aspen, but this does not affect the rest of the flowsheet since only the product distribution is required. Sand is defined as an inert to prevent unintended participation in the yield model. The reactor operating conditions are fixed at 480 °C with a 0.3 atm pressure drop (entered as -0.3 atm in Aspen).

A mixer is placed immediately after the reactor to combine (1) the reactor effluent, (2) the separated moisture stream, and (3) the fluidizing-gas stream. This mixer is intentionally a “fake” piece of equipment required only to prevent Aspen from forcing the fluidizing gas to participate in the RYIELD decomposition. Because the mixer cools the stream (the fluidizing gas enters at 375 °C and the moisture at 250 °F), a heater is added after the mixer to raise the mixed stream back to 480 °C prior to entering the real cyclone that separates solids from gases. As with all heaters and heat exchangers in the process, no pressure drop is assumed unless explicitly required.

After being reheated to 480 °C, the mixed stream enters an SSPLIT cyclone, where all NC and CI solids are separated from the MIXED-phase vapors; the vapor stream is directed upward toward the quench-absorber section, while the separated solids flow downward into the char-combustor section for further processing.

In the char-combustor hierarchy, a calculator block transfers the sand outlet temperature from the combustor loop back to the sand-inlet stream feeding the pyrolysis reactor. A design specification then adjusts the heater on the sand stream so that this temperature is varied until the pyrolysis reactor operates adiabatically. These details are discussed later in the design-specification section, but they are mentioned here to show how the sand-temperature loop connects the two hierarchies.

methodology for modeling the Quench Absorber

After leaving the solids–gas separator, the vapor stream enters the quench-absorption section, beginning with a heat exchanger that precools the gas prior to absorption. This exchanger is modeled as a simple HeatX block in which the hot pyrolysis vapors transfer heat to the recycled light-gas stream exiting the top of the quench absorber. A zero pressure drop is assumed across this exchanger. The exchanger is specified by setting the cold-stream outlet temperature to 375 °C, because this recycled gas ultimately represents the fluidizing gas entering the pyrolysis reactor. The cold side is the recycled gas stream since its inlet temperature is significantly lower than that of the hot vapors.

Because the Quench Absorber itself was modeled as a SEP2 unit, Aspen does not allow direct specification of absorber operating temperature. Therefore, a heater placed immediately before the SEP2 is used to impose the absorber conditions. This heater was set to 45 °C and 1.7 atm, representing the provided absorber operating conditions. This is a major but reasonable assumption, as the problem statement gives no heater design details. This assumption significantly influences the number of cooling stages required downstream and ultimately determines the amount of LP steam, HP steam, and cooling-water duty generated in later heat-recovery units. These effects are discussed in more detail in the Design Spec and Calculator Block section.

The SEP2 is configured to split components according to the recoveries described verbally by the professor during lecture. All light non-keys (LNKs)— H_2 , N_2 , CO , O_2 , CH_4 , C_2H_4 , CO_2 , SO_2 , and NO_2 —are assigned 100% recovery in the top outlet and represent the recycled light-gas stream. All heavy non-keys (HNKs)—the organics—are assigned 100% recovery in the bottom outlet, which forms the crude liquid product stream. Two components have partial recoveries prescribed explicitly: acetone, which is the light key with 96% recovery in the vapor phase, and water, the heavy key with 20% recovery in the vapor phase. These values were directly supplied by the professor. All other organics (acetic acid, m-cresol, guaiacol, levoglucosan, and furfural) are recovered completely in the liquid phase.

Because the recoveries are specified explicitly, there is no need for the pump + cooler internal recycle loop shown in the assignment's schematic to enhance absorber performance. Instead, the SEP2 alone sets the vapor-liquid distribution.

The bottoms stream from the SEP2 enters a decanter, which separates the aqueous phase from the organic bio-oil. The decanter is operated with zero duty and no pressure drop. The second liquid phase is identified using water as the key component with an efficiency of 100%. Despite specifying perfect phase-split efficiency, a small amount of trace organics (≈ 0.15 wt%) remains in the aqueous phase, while the remainder reports to the organic bio-oil product.

The top vapor stream from the SEP2 is first heated before compression. Because the light-gas mixture at $45^\circ C$ and 1.7 atm lies below its dew point, the compressor would generate errors if fed directly. Through trial-and-error dew-point checking, the dew point was estimated at approximately $65^\circ C$, which is the outlet temperature imposed in the heater preceding the compressor. This heater is visually minimized and disguised on the main flowsheet since it serves only to avoid compressor convergence errors.

The heated vapor stream then enters an isentropic compressor, which raises the pressure to 2.31 atm to match the fluidizing-gas pressure entering the pyrolysis reactor. The outlet of the compressor is fed into another heater, which cools it to $80^\circ C$ to meet the bleed gas temperature specification provided in the problem statement. Like before, this heater is minimized on the main flow sheet due to it only serving to meet one specification and its small cooling rate. The outlet of the heater is split using a Flow Splitter (FSPLIT) into two streams:

1. Fluidizing-gas stream toward the pyrolysis reactor (not actually closed into a loop, but treated as such conceptually). The FSPLIT is specified to deliver 35,000 kg/hr, following the fluidization-gas-to-sand ratio given in the assignment.
2. Bleed-gas stream toward the char combustor. Although the assignment nominally specifies 13,000 kg/hr, the model yields approximately 12,500 kg/hr, the highest

value achievable without violating other assignment constraints or underfeeding the pyrolysis fluidizing gas.

It is important to note that both split streams contain some H_2 originating from pyrolysis, meaning their compositions differ slightly from the pre-loop molar compositions listed in the assignment. This discrepancy is unavoidable since Aspen cannot remove hydrogen selectively while maintaining the required mass-flow constraints.

The bleed-gas stream leaving the splitter is then fed directly into the char combustor, which begins the next major section of the flowsheet.

methodology for modeling the Char Combustor

After the solids–gases split leaving the pyrolysis section, the sand + char mixture enters the char-combustor hierarchy. This hierarchy contains two sequential RSTOIC reactors that together model (1) char decomposition into elemental species and (2) full combustion with bleed gas and air.

The first RSTOIC block (R-300A) receives char and sand from the cyclone. Char is entered as the single reactant with a stoichiometric coefficient of -1 , and the products— H_2 , O_2 , N_2 , elemental carbon (C-solid), sulfur, and ash—are each assigned an initial stoichiometric coefficient of 1. These coefficients are placeholders only; the true stoichiometric coefficients are later calculated from (1) the ultimate analysis of char (assumed to be the same as dry biomass) and (2) the molecular weights of the product species. These corrected values are imported automatically using a calculator block described later in the report.

Ash composition is defined in its component attributes as 100% ash on a mass basis (an assumption taken from lecture). The reactor is modeled as adiabatic with an operating pressure of 1.12 atm. Sand is inert in this reactor and simply passes through.

The effluent from this first RSTOIC flows to a second RSTOIC (R-300B) that models full combustion. The reactor receives (1) the char-decomposition products, (2) the bleed-gas stream from the quench-absorber recycle loop, and (3) preheated air. To avoid convergence issues in Aspen, NO is specified as the primary combustion product instead of NO_2 . The operating pressure (1.12 atm) and adiabatic assumption are kept consistent with the first reactor.

A cyclone immediately downstream separates C-solid sand from the ash + gas mixture. This cyclone is placed inside the hierarchy so that the main flowsheet shows only the single ash/fines separator depicted in the assignment PFD. The separated sand is routed to a heater outside the hierarchy, where its temperature is adjusted using a design specification

so that the pyrolysis reactor is adiabatic. The heater pressure is manually set to 2.3 atm to match the sand pyrolysis reactor inlet since no unit exists in Aspen to directly control solid-phase pressure.

The gas-phase outlet of the hierarchy (ash + fines + mixed gas) passes to the main flowsheet cyclone, which separates the ash product stream and a gas stream containing N_2 , CO_2 , O_2 , SO_2 , NO_x , H_2O , and minor H_2/CO from upstream sources. This gas enters a heater set to 250 °C. The heater is included to ensure that the inlet temperature to the air-preheat exchanger remains physically reasonable, as the mixture would otherwise exceed 850–900 °C leaving the combustor.

The cooled gas is fed to a heat exchanger (E-301) where ambient air acts as the cold stream. The exchanger specification is the cold-stream outlet temperature, set to 100 °C, which yields an appropriate air-preheat for combustion. The air stream is supplied at 25 °C and 1 atm, with composition assumed as 79% N_2 / 21% O_2 .

The preheated air is sent to the second RSTOIC combustor, since no combustion can occur without O_2 present. To satisfy the requirement that the combustor effluent contains 2 mol% O_2 , a design specification adjusts the inlet air flowrate until this target oxygen fraction is achieved. Keeping O_2 in slight excess avoids failure of the RSTOIC block when insufficient oxygen is supplied for complete combustion, while avoiding unnecessary excess air.

This concludes the char-combustor section, after which the report proceeds to the detailed explanation of the calculator blocks and design specifications that support stoichiometry correction, adiabatic constraints, and sand-loop temperature matching, as well as reporting and calculating deliverables.

Design Specifications

Several design specifications were required to ensure proper operation of the flowsheet, satisfy assignment constraints, and determine steam generation across all heat-exchange stages. These design specifications directly manipulate inlet stream conditions or heat-exchanger cooling-water flowrates so that key temperature, pressure, and composition targets are met.

1. DS-SAND — Adiabatic Pyrolysis Reactor

The first design specification, DS-SAND, forces the pyrolysis reactor to operate adiabatically.

Because the sand loop is not closed in Aspen, the temperature of regenerated sand must be controlled externally. DS-SAND accomplishes this by:

- **Target:** Set the pyrolysis reactor heat duty (QCALC) to 0 kW.

- **Varied variable:** The outlet temperature of heater E-302, which cools the regenerated sand after the combustor.
- **Mechanism:** A calculator block transfers the E-302 outlet temperature into the *sand-to-reactor* inlet stream.
DS-SAND then adjusts this temperature until the reactor becomes perfectly adiabatic.

This ensures the sand supplies exactly the heat required to drive the endothermic pyrolysis reactions.

2. DS-AIR — Ensuring Adequate Oxygen for Complete Combustion

The second design specification, DS-AIR, adjusts the combustion-air flowrate entering the second RSTOIC reactor (char combustor).

- **Target:** Achieve 2 mol% O₂ in the combustor effluent.
- **Varied variable:** The inlet molar flow of the AIR stream.

Combustion cannot occur without O₂, and a small excess is required to ensure complete conversion of char, CO, H₂, and other combustibles. The assignment specifies 2 mol% excess O₂, which is used here as a reasonable and safe margin. DS-AIR ensures the RSTOIC block does not fail due to oxygen starvation while avoiding excessive air consumption.

3. DS-E30X — Heat Exchangers for Steam Generation

The three heaters on the main flowsheet (E-401 upstream of the quench absorber, E-300 downstream of the char combustor, and E-302 on the hot-sand loop) were each placed inside a hierarchy and re-modeled as a series of simple heat exchangers. These internal exchangers do not change the overall flowsheet topology; they are used only to convert the sensible cooling of the hot streams into high-pressure (HP) steam, low-pressure (LP) steam, or warmed cooling water. Within each hierarchy, the minimum temperature approach was set to 8 °C so Aspen would not conflict with the 10 °C temperature differences imposed through the design specs below.

For each internal exchanger, a design specification adjusts the cooling-water inlet flowrate so that the utility outlet temperature matches the desired saturation or outlet temperature. The heat exchanger specification is set as the hot stream outlet temperature. This provides a consistent basis for computing the steam generated over each temperature interval.

E-401 hierarchy (three stages: HP steam, LP steam, CW heating)

The gas leaving the main heater E-401 must ultimately enter the quench absorber at 45 °C and 1.7 atm. Inside the hierarchy, this cooling from roughly 380 °C down to 45 °C is split into three exchangers:

1. Stage 1 (HP steam):

- Hot stream cools from ~380 °C to 260 °C.
- Cold side produces HP steam.
- Design spec: vary cooling-water inlet so HP-steam outlet temperature is 250 °C.
- The resulting HP-steam generation is on the order of slightly under 8 MMscfd.

2. Stage 2 (LP steam):

- Hot stream cools from 260 °C to 135 °C.
- Cold side produces LP steam.
- Design spec: adjust cooling-water inlet so LP-steam outlet temperature is 125 °C.
- LP-steam production is slightly higher than the HP-steam from Stage 1 but still modest, also around 8 MMscfd.

3. Stage 3 (cooling water only):

- Hot stream cools from 135 °C to 45 °C.
- No steam is generated; only cooling water is heated.
- Design spec: set cooling-water outlet temperature to 35 °C by changing inlet flow (inlet fixed at 25 °C, 1 atm).
- Because of the relatively small temperature driving force at low temperature and the large duty required to cool to 45 °C, the cooling-water flowrate is extremely large—on the order of 1000+ MMscfd.

Overall, E-401 generates only small amounts of HP and LP steam compared with the very large cooling-water duty required to bring the vapors all the way down to quench-absorber conditions.

E-300 hierarchy (two stages: HP steam, LP steam)

The flue-gas stream entering E-300 is at approximately 860 °C. On the main flowsheet its outlet is fixed at 250 °C, so only HP and LP steam are generated inside the hierarchy:

1. Stage 1 (HP steam):

- Hot stream cools from ~860 °C to 260 °C.

- Generates HP steam.
- Design spec: vary cooling-water inlet so HP-steam outlet temperature is 250 °C.
- This stage produces a moderate amount of HP steam, roughly 30 MMscfd, reflecting the large high-temperature duty available across this interval.

2. Stage 2 (LP steam):

- Hot stream cools from 260 °C to 250 °C.
- Treated as LP-steam generation.
- Design spec: adjust cooling-water inlet so LP-steam outlet is 125 °C.
- Because of the very narrow temperature window, the resulting LP-steam flow is very small, on the order of 0.5 MMscfd.

Thus E-300 is an intermediate contributor to steam production: it provides a noticeable amount of HP steam, but LP steam from this exchanger is almost negligible.

E-302 hierarchy (single stage: HP steam)

The hot sand leaving the char combustor enters E-302 at roughly 529 °C and is cooled to 250 °C before being “sent back” to the pyrolysis reactor. Since this entire cooling range lies above the HP-steam level, only one exchanger is needed inside the hierarchy:

- **Single stage (HP steam):**
 - Hot sand cools from ~529 °C to 250 °C.
 - Cold side produces HP steam.
 - Design spec: adjust cooling-water inlet so HP-steam outlet temperature is 250 °C.
 - Because the sand stream is large and very hot, this stage generates the largest single contribution of HP steam, about 60 MMscfd.

Aspen flags the E-302 exchanger because it involves solid–liquid heat transfer, but the warning is harmless and does not affect the steam-generation computation.

Comparison of steam-generation trends

Comparing the three sections:

- **E-401** produces small amounts of HP and LP steam (≈8 MMscfd each) but an enormous amount of warm cooling water (1000+ MMscfd), because it cools already-partially-cooled vapors all the way down to 45 °C, where steam generation is no longer thermodynamically favorable and most of the duty is in the low-temperature range. This can be mitigated if another specification is set for this

heater that does not bring its outlet temperature below the LP steam temperature range.

- **E-300** generates a moderate HP-steam flow (~30 MMscfd) and almost negligible LP steam (~0.5 MMscfd). This reflects the high inlet temperature of the flue gases but a relatively modest temperature span available for LP-steam generation.
- **E-302** dominates HP-steam production with roughly 60 MMscfd, since it cools a large mass of hot sand entirely within the HP-steam temperature range, with no low-temperature tail that would require cooling water instead.

These trends are physically reasonable: the largest steam production comes from cooling the hottest and heaviest stream (the circulating sand), intermediate steam production comes from the high-temperature flue gases, and the quench-absorber pre-cooler mostly acts as a cooling-water sink, with only modest steam generation because most of its duty occurs at relatively low temperatures.

Calculator Blocks

Four calculator blocks were used in the simulation, each serving a distinct purpose: temperature transfer for the sand loop, steam-flow aggregation, stoichiometric coefficient generation, and carbon-yield calculations. Their functions and internal logic are described below.

C-TEMP — Temperature Transfer for the Sand Loop

The C-TEMP calculator block transfers the sand temperature leaving the char-combustor heater (E-302) to two locations:

1. **The hot-sand inlet to the pyrolysis reactor**, allowing the design specification *DS-SAND* to vary the sand temperature until the pyrolysis reactor becomes adiabatic ($Q_{\text{CALC}} = 0$).
2. **The heat-exchanger surrogate for E-302 inside the steam-generation hierarchy**, ensuring that the HP-steam calculation updates correctly even when the E-302 temperature changes externally.

This is necessary because the sand loop is not physically closed in the flowsheet; therefore Aspen will not automatically propagate temperature between the units unless it is explicitly transferred by a calculator block.

C-STEAM — Summation of Steam Production

The C-STEAM calculator block simply aggregates all steam produced across the model.

Inputs:

- HP Steam Streams: HPSTEAM, HPSTEAM2, HPSTEAM3
- LP Steam Streams: LPSTEAM, LPSTEAM2

The calculator block adds their flow rates (in MMSCFD) and outputs:

- Total HP steam (MMSCFD)
- Total LP steam (MMSCFD)

This block is purely arithmetic and does not feed back into the flowsheet—its outputs are exported only for reporting in the KPIs custom table.

C-RXN — Stoichiometric Coefficients for Char Conversion

The first RSTOIC in the char-combustor hierarchy (R-300A) converts Char into its elemental products: H₂, O₂, N₂, Carbon (C solid), Sulfur, and Ash.

However, the assignment does not provide stoichiometric coefficients, so they must be computed from the ultimate analysis of Char.

Since Char is treated as bone-dry, moisture does not appear in the formula. The stoichiometric coefficient for each product is calculated as:

$$v_i = \frac{\text{ULTANAL}(i)}{100 \times \text{MW}_i}$$

For example, if the ultimate analysis gives 49.66 wt% carbon:

$$v_c = \frac{49.66}{100 \times 12.011}$$

These calculations are performed for C, H, N, S, O, using the imported molecular weights for each elemental or pseudo-elemental product.

The resulting coefficients are exported directly into R-300A's stoichiometric table, replacing the initial unity guesses.

Carbon-Yield Calculator — Bio-Oil, Waste-Water, and Flue-Gas Carbon Yields

The final calculator block computes:

- Carbon yield to bio-oil (%)
- Carbon yield to aqueous phase (%)
- Carbon yield to flue gas (%)

This is done by comparing the mass flow of carbon entering with biomass to the mass flow of carbon leaving in each product stream.

A. Carbon entering with dry biomass

The carbon entering the pyrolysis reactor is:

$$CFLOW = \dot{m}_{\text{biomass}} \times \frac{ULTANAL(C)}{100}$$

This value is exported for later use.

B. Mass Fraction of Carbon in Each Organic Compound

To determine carbon flow in each product stream, the mass fraction of carbon in each compound must be known.

These were calculated manually in Excel from molecular weights imported from Aspen and entered as constants into the calculator block.

Example Calculation (CO₂):

$$X_{C,CO_2} = \frac{12.011}{12.011 + 2(15.999)} = 0.2729$$

As another example, for LVG (C₆H₁₀O₅):

$$X_{C,LVG} = \frac{6(12.011)}{MW_{LVG}}$$

Mass-fraction constants entered for all carbon-bearing components:

- ACETONE
- AACID
- FURFURAL
- MCRESOL
- GUAICOL
- LVG
- CO₂

C. Bio-Oil Carbon Flow

Bio-oil consists mostly of organics, with very trace amounts of water.

For each component of the six organics:

$$\dot{m}_{C,i} = \dot{m}_i \times X_{C,i}$$

Summing:

$$\dot{m}_{C,\text{bio-oil}} = \sum_i \dot{m}_{C,i}$$

Yield:

$$\text{Bio-Oil Carbon Yield (\%)} = 100 \times \frac{\dot{m}_{C,\text{bio-oil}}}{CFLOW}$$

D. Waste-Water Carbon Flow

Opposite to Bio-Oil, the wastewater contains water plus trace organics. The same calculation as above is performed but with much smaller organic flows.

E. Flue-Gas Carbon Flow

Only CO₂ contains carbon:

$$\dot{m}_{C,\text{flue}} = \dot{m}_{CO_2} \times X_{C,CO_2}$$

Yield:

$$\text{Flue-Gas Carbon Yield (\%)} = 100 \times \frac{\dot{m}_{C,\text{flue}}}{\text{CFLOW}}$$

The calculated yields, as well as compressor power and steam generation rates (HP, LP) are reported in a custom table, which can be seen in the next section.

Section #4: Miscellaneous results

The results of the simulation are summarized in the table below, which reports the compressor power, the HP- and LP-steam generation rates, and the carbon-based yields to bio-oil, waste water, and flue gas. The bio-oil yield is the largest, reflecting that a significant fraction of the condensable carbon species formed in the pyrolysis reactor are recovered in the organic liquid phase of the quench-absorber and decanter system. A substantial portion of carbon appears in the flue gas, which is expected since combustion of char and the bleed-gas stream converts a large share of the remaining carbon to CO₂. Only a very small amount reports to the waste-water phase because the aqueous layer contains only dilute concentrations of partially soluble oxygenates. The small remainder of carbon is attributable to trace amounts in solid streams and minor numerical rounding.

The steam-generation results are strongly influenced by the temperature ranges available in each heater. Most of the cooling duties occur well above the HP-steam saturation temperature, so a large fraction of the available sensible heat naturally converts to HP steam. The sand cooler in particular produces the most HP steam because it cools a large, high-mass-flow solid stream from several hundred degrees above the HP-steam level, allowing a substantial amount of energy recovery. The combustor outlet also contributes a moderate amount of HP steam due to its high inlet temperature relative to the 250 °C cutoff. In contrast, the LP-steam window is narrow, both in terms of temperature range and the specific locations where the gases cross into the LP-steam region. As a result, only a small portion of the total duty falls within the 125 °C zone where LP steam can form. Once the gas temperatures fall below the LP-steam range, the remaining heat is too low-grade to generate steam and must instead be rejected to cooling water. Overall, the high proportion

of HP steam reflects both the large thermal duties in the high-temperature zones and the limited opportunities to generate LP steam as the process streams approach quench-absorber conditions.

KPIs			
	Name	Units	Value
▶	Power	kW	426.331
▶	LP-STEAM	MMSCFD	9.52958
▶	HP-STEAM	MMSCFD	101.859
▶	Bio-Oil yield (%)	UNITLESS	62.2802
▶	Waste Water yield (%)	UNITLESS	0.151448
▶	Flue Gas yield (%)	UNITLESS	32.4684

Figure 5: Custom table

Section #5: References

Most information used to solve this problem was derived from the materials provided in the problem description or from lessons taught by Dr. Michael Caracotsios. No external sources or research articles were consulted. Aspen Plus V14 was used to perform the simulation and costing calculations, while ChatGPT was utilized to assist with navigating Aspen and refining the clarity and formatting of this report.

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